

Reproducible Consistency Training via Improved Self-Distillation Guidance

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Abstract

Consistency models have been proposed for fast generative modeling, achieving competitive results compared to diffusion and flow matching models. However, these methods often rely on heuristics to mitigate training instability, which in turn limits reproducibility and scalability. To address this limitation, we propose the generalized flow map framework, unifying recent consistency-based methods under a common perspective. Within this framework, we investigate the suboptimality of existing approaches and identify two key factors for reproducibility: time-condition relaxation and marginal velocity guidance. To incorporate these, we leverage self-distillation to guide consistency models along the marginal velocity. We further propose improved Self-Distillation (iSD) by exploring the design space of flow maps, thereby reducing reliance on heuristics. Our formulation naturally extends to classifier-free guidance, achieving two-step generation with an FID of 9.70 on ImageNet 256×256 . iSD provides a theoretical and empirical foundation for reproducible consistency training.

1. Introduction

Diffusion [14, 33–35] and flow matching models [19, 21] have achieved remarkable performance across a wide range of applications. This progress stems from flow-based modeling and multi-step inference, but is limited by multiple network evaluations during generation. To address this limitation, several works have explored improving sampling efficiency [29, 39, 42, 43, 46], but they rely on additional distillation stages or auxiliary networks to achieve fewer-step generation, thereby introducing additional training cost.

Consistency Models [36] and its variants [8, 10, 23, 32, 38, 40] have been proposed for training from scratch in few-step generation. Earlier studies on consistency models have often suffered from training instability, which led subsequent works to focus on stabilization by introducing various heuristics. However, the reliance on these complex techniques has reduced reproducibility.

In this work, we aim to develop a simplified and re-

producible few-step generative model based on consistency methods. We begin by analyzing existing approaches: (i) we propose a generalized flow map framework that covers various design choices of recent consistency-based methods, (ii) we unify these approaches under our framework, providing a theoretical basis for analyzing suboptimality, and (iii) we show that recent methods do not guarantee convergence to the generator along the marginal velocity field, due to suboptimality and instability of their objectives.

Motivated by these observations, we hypothesize that suboptimality and instability undermine the reproducibility of consistency training. From our unified perspective, we identify two key factors for reproducible training: time-condition relaxation and marginal velocity guidance. To incorporate these factors, we leverage our generalized flow map formulation with self-distillation. Some prior work [15, 31] have attempted to address these issues by reducing the loss variance, thereby resolving them indirectly. In contrast, self-distillation [3] was proposed to guarantee convergence to the marginal flow directly, but it relies on heuristics to stabilize training. By exploring the design space of flow map models, we propose improved Self-Distillation (iSD), which further reduces the reliance on heuristics. Moreover, we extend iSD to classifier-free guidance, achieving additional performance gains.

Our iSD demonstrates an FID [12] of 9.70 and Inception Score [30] (IS) of 246.85 for two-step generation on ImageNet 256×256 [6]. The reproducibility of iSD is validated across multiple random initializations by measuring the standard deviation of quantitative metrics, achieving 0.05 for FID and 0.50 for IS. These demonstrate improved reproducibility compared to consistency training, as training can be performed from scratch with reduced heuristics.

Contribution. (i) We extend the flow map framework to cover various design choices, unifying recent consistency-based approaches within it (Sec. 3.2). (ii) We prove the sub-optimal convergence and instability of gradient dynamics in recent consistency-based methods, showing that they undermine reproducibility and training stability (Sec. 3.3). (iii) To address these issues, we leverage the self-distillation, which guarantees convergence to the marginal velocity field

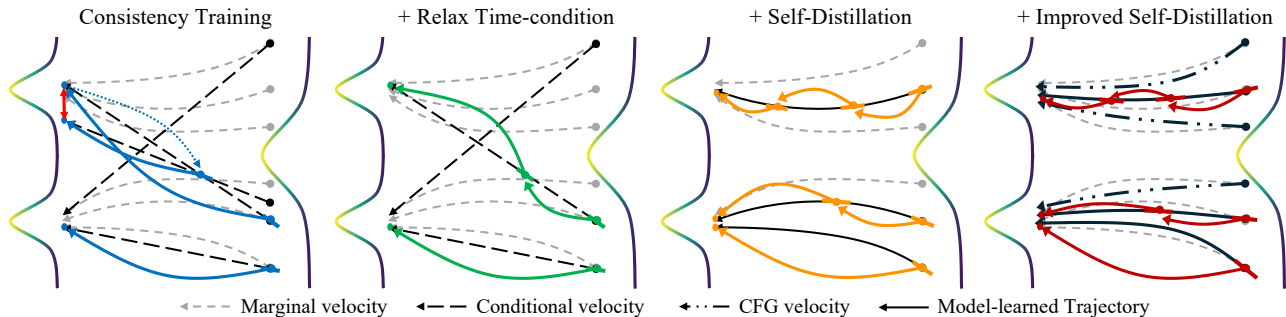


Figure 1. **From Consistency Training to improved Self-Distillation.** Consistency training learns mapping along the conditional velocity field, often suffering from training instability and reproducibility issues. Relaxing the time condition mitigates this instability, and applying self-distillation guarantees the mapping along the marginal velocity field. By exploring design spaces and incorporating classifier-free guidance, improved Self-Distillation (iSD) reduces reliance on training heuristics and further improves performance and reproducibility.

(Sec. 4). (iv) We generalize the self-distillation to incorporate recent design choices and further extend it to classifier-free guidance, which we term *improved Self-Distillation* (iSD, Sec. 4). (v) We explore the design space and present the best choices for iSD. (Sec. 5).

2. Related Work

Diffusion and Flow Matching Models. Diffusion models [14, 33, 35] and flow matching models [1, 2, 4, 21] are generative models that gradually transform a tractable noise distribution into the data distribution. These models have achieved remarkable progress in high-fidelity generation [7, 24, 26, 27]. However, their reliance on a multi-step sampling requires substantial computational resources.

Few-step Generation. Several work have explored improving sampling efficiency of diffusion models [27, 29, 39]. These approaches aim to distill pretrained diffusion models into fewer-step generators, adopt GANs, or leverage VAEs to reduce input size. In parallel, distribution matching distillation methods [42, 43, 46] have been proposed to construct one-step generators by tracking the generator’s score. However, both approaches rely on additional distillation stages or auxiliary networks, which increase training cost.

Consistency Models. Consistency Models [36] are designed to predict a sample directly from any point along a flow trajectory. Both distillation and training methods have been proposed, but training from scratch is known to be unstable. Several studies have introduced heuristics to stabilize training, including initialization, improved objectives, and progressive training schemes [10, 23, 32]. Other studies have identified a gap between distillation and training objectives [4, 15, 31], which can lead to high loss variance and suboptimal convergence when training from scratch. To mitigate this, network-induced couplings were introduced to reduce loss variance, addressing the issue indirectly. In

contrast, self-distillation was proposed to guarantee convergence directly, while dependent on heuristics.

Unified Framework. Recently, several studies have aimed to unify flow matching and consistency models. UCGM [38] introduced a framework that integrates both paradigms, but it does not account for the relaxed constraint of mapping points. The flow map framework [4, 17] presented a mathematical framework for consistency models, defining a model that learns *flow maps* as mappings between any two points on the same trajectory. However, it does not aim to unify recent work or explore its design spaces.

3. Suboptimality of Direct Flow Map Matching

In this section, we revisit the flow map framework [4, 17] and extend it to interpret recent consistency-based approaches. We observe that most methods learn conditional velocity fields, which do not guarantee convergence to the marginal velocity field. Such suboptimal flow maps may lead to trajectory crossings, non-injective mappings, or severe reproducibility issues.

3.1. Preliminaries

Given a training dataset X with underlying distribution p_X , flow matching models are trained to match the velocity fields of continuous flows, starting from a tractable distribution p_Z . Prior work constructs such flows via an interpolation, $x_t = \alpha_t x + \sigma_t z$, where $x \sim p_X$ and $z \sim p_Z$. A linear interpolation [19, 21] and a trigonometric interpolation [1, 2, 23] are the widely adopted choices.

We assume that α_t and σ_t are continuous and monotone with boundary conditions $\alpha_0 = \sigma_T = 1$ and $\alpha_T = \sigma_0 = 0$ for $t \in [0, T]$. Under bounded first- and second-derivative, the marginal distribution induced by the flow, ρ_t , satisfies $\rho_0 = p_X$ and $\rho_T = p_Z$. To ensure convergence of the consistency objective, we propose an additional assumption:

$\alpha_t \sigma'_t - \sigma_t \alpha'_t = \nu \neq 0$ for all $t \in [0, T]$, where ν is a constant (see Appendix A.4). Notably, both linear and trigonometric interpolations satisfy this condition with $\nu = 1$.

With the constructed flow, the flow matching models optimize the squared error between the conditional velocity $v_t(x_t|x) = \alpha'_t x + \sigma'_t z$ and a velocity network $F_\theta(x_t; t)$:

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{x,z,t} [\|F_\theta(x_t; t) - v_t(x_t|x)\|_2^2]. \quad (1)$$

Conditional flow matching $\mathcal{L}_{\text{CFM}}$ converges to the flow induced by the marginal velocity $v_t^*(x_t) = \mathbb{E}_{x|x_t} [v(x_t|x)]$. If $v_t^*(x)$ is Lipschitz continuous in both t and x , the ODE $dx_t = v_t^*(x_t)dt$ has a unique solution and satisfies the continuity equation for ρ_t [19]. However, we observe that diffusion and flow matching models can suffer from *mean collapse*, in which one-step samples collapse to the mean of the data distribution (proof in Appendix A.1).

3.2. Generalized Flow Map Matching

We begin by introducing a flow map with Eulerian distillation following prior work, and generalize these formulations to unify recent consistency-based methods. Within our extended framework, we interpret recent methods and provide a theoretical basis for analyzing their suboptimality.

Flow Map. From the flows defined by an interpolation, our goal is to draw samples from the target distribution in a few sampling steps. To achieve this, we define a *flow map* $f_{t,s}(x_t)$ as a mapping between two points x_t and x_s ($s < t$) on the same trajectory, using the marginal velocity $v_t^*(x_t)$:

$$f_{t,s}(x_t) = x_t + \int_t^s v_\tau^*(x_\tau) d\tau = x_s. \quad (2)$$

If the marginal velocity is Lipschitz continuous, the flow map is well-defined, injective, and thus *free from the mean collapse problem* (proof in Appendix A.2).

Training Flow Map. Since the flow map is defined as an integral, direct supervision from scratch is challenging. Assuming a teacher flow matching network Φ , we can generate training data using an ODE solver, enabling direct supervision in a distillation manner. This procedure reduces to the one-step distillation of rectified flows [21] for fixed $t = 1$ and $s = 0$. However, it requires a data generation process with substantial computational resources.

To address this limitation, many recent studies adopt consistency training [36], which can be derived from the Eulerian equation (proof in Appendix A.3; see also [4]).

Proposition 1. (Eulerian Equation) *Given the flow induced by a Lipschitz continuous marginal velocity $v_t^*(x_t)$, the flow map $f_{t,s}$ is the unique solution of the Eulerian equation:*

$$\partial_t f_{t,s}(x) + v_t^*(x_t) \cdot \nabla_x f_{t,s}(x) = 0, \quad (3)$$

when f is continuous in x, t, s , Lipschitz continuous in x , and satisfies the boundary $f_{t,t}(x) = x$.

To facilitate training of a flow map network $f_\theta(x_t; t, s)$ using the Eulerian equation, the training objective can be formulated as a squared minimization problem, referred to as *Eulerian distillation*:

$$\mathcal{L}_{\text{ED}} = \mathbb{E}_{x,z,t,s} [\|\partial_t f_\theta + v_t^* \cdot \nabla_x f_\theta\|_2^2], \quad (4)$$

where $f_\theta = f_\theta(x_t; t, s)$ and $v_t^* = v_t^*(x_t)$. When $s = 0$, Eulerian distillation reduces in value to continuous time consistency training objective [36] (details in Appendix A.5). However, their gradient dynamics differ, which affects the training stability, as further discussed in Proposition 4.

Since the objective involves a Jacobian-vector product, its optimization requires computing a Hessian, introducing computational overhead. To address this, sCT [23] reformulated the consistency training objective with a stop-gradient operation, while keeping the gradients identical:

$$\mathcal{L}_{\text{CT}} = \mathbb{E}_{x,z,t} \left[\left\| f_\theta - f_{\theta-} + \frac{df_{\theta-}}{dt} \right\|_2^2 \right], \quad (5)$$

where $f_{\theta-}$ denotes the gradient-detached network. Using Eq. (5), the JVP operation is detached from the gradient flow, and the update step requires two backward passes but avoids Hessian computation.

Generalized Flow Map. To interpret recent approaches as flow maps, we propose a generalized formulation of f_θ as a one-step Euler solution with a pseudo-velocity network F_θ :

$$f_\theta(x_t; t, s) = \nu^{-1}(A'_{t,s}x_t - A_{t,s}F_\theta(x_t; t, s)), \quad (6)$$

where $A_{t,s} = \sigma_t \alpha_s - \sigma_s \alpha_t$. This formulation satisfies the boundary condition of Proposition 1. For a linear trajectory, it reduces to the form widely adopted in flow map studies [4, 9, 28]. At $s = 0$, it simplifies to the common setting in consistency models [38, 40]. For a trigonometric trajectory with $s = 0$, we obtain the form introduced by [23].

By generalizing the formulations of flow maps, we propose instantiations of these methods under a unified perspective (proof in Appendix A.5).

Proposition 2. (Interpretation of Recent Methods) *Recent consistency models can be interpreted as instances of the flow map, trained with the generalized Eulerian equation:*

$$\partial_t f_\theta(x_t; t, s) + \tau_t(x_t, x) \cdot \nabla_x f_\theta(x_t; t, s) = 0, \quad (7)$$

where x_t is given by the Interpolant, τ_t by the Trajectory, and t, s by the Timestep, as summarized in Tab. 1.

To be consistent with the generalized Eulerian equation, we extend two objectives for the guiding trajectory $\tau_t = \tau_t(x_t, x)$. Since it is not generally Lipschitz, it no longer guarantees the convergence to the marginal flow map.

$$\mathcal{L}_{\text{ED}} = \mathbb{E} [\|\partial_t f_\theta + \tau_t \cdot \nabla_x f_\theta\|_2^2], \quad (8)$$

$$\mathcal{L}_{\text{CT}} = \mathbb{E} [\|f_\theta - f_{\theta-} + \partial_t f_{\theta-} + \tau_t \cdot \nabla_x f_{\theta-}\|_2^2]. \quad (9)$$

Table 1. **Instantiation of the flow map framework** for each consistency-based generative model. *Interp.* denotes the interpolant, *Traj.* the guiding trajectory, and *Time.* the timestep conditions. Note that $\Phi(x_t; t)$ is the teacher network and $F_\theta(x_t; t, s)$ is the training network.

Model	Interp. x_t	Traj. τ_t	Time.	JVP	Loss
Distillation-based Methods					
FMM-EMD [4]	Linear	$\Phi(x_t; t)$	$s < t$	Exact	$\mathcal{L}_{ED}$
AYF-EMD [28]	Linear	$\Phi(x_t; t)$	$s < t$	Exact	$\mathcal{L}_{CT}$
sCD [23]	Trig.	$\Phi(x_t; t)$	$s = 0$	Exact	$\mathcal{L}_{CT}$
Consistency Training Methods					
MeanFlow [9]	Linear	$v_t(x_t x)$	$s < t$	Exact	$\mathcal{L}_{CT}$
ConsistencyFM [40]	Linear	$v_t(x_t x)$	$s = 0$	Approx.	$\mathcal{L}_{CT}$
sCT [23]	Trig.	$v_t(x_t x)$	$s = 0$	Exact	$\mathcal{L}_{CT}$
UCGM [38]	Arbitrary	$v_t(x_t x)$	$s = 0$	Approx.	$\mathcal{L}_{CT}$
Shortcut Model [8]	Linear	$F_\theta(x_t; t, t)$	$s < t$	Approx.	$\mathcal{L}_{CTM}$
Self-Distillation Methods					
Self-Distillation [3]	Linear	$F_\theta(x_t; t, t)$	$s < t$	Exact	$\mathcal{L}_{SD}$
iSD (Ours)	Arbitrary	$F_\theta(x_t; t, t)$	$s < t$	Approx.	$\mathcal{L}_{SD-R}$

We observe several design choices from these instantiations. Among them, we observe that the guiding trajectory and time condition primarily influence training stability and reproducibility. In the following subsection, we analyze how these factors contribute to such issues and demonstrate the most effective settings in the experimental section.

3.3. Suboptimality and Instability

Most consistency training approaches learn flow maps guided by the conditional velocity. We denote Eulerian distillation under a conditional velocity field as *direct training*.

$$\mathcal{L}_{DT} = \mathbb{E}_{x,z,t,s} [\|\partial_t f_\theta + v(x_t|x) \cdot \nabla_x f_\theta\|_2^2]. \quad (10)$$

However, the direct training does not guarantee convergence to the flow map along the marginal velocity, due to the gap between conditional and marginal velocity guidance (proof in Appendix A.6; cf. [4]).

Proposition 3. (*Suboptimality of direct training*) *The gap between Eulerian distillation $\mathcal{L}_{ED}$ along the marginal velocity and direct training $\mathcal{L}_{DT}$ is given by*

$$\mathcal{L}_{DT} - \mathcal{L}_{ED} = \mathbb{E}_{x,z,t,s} [\text{Var}_{x|x_t} [\Delta v \cdot \nabla_x f_\theta]], \quad (11)$$

where $\Delta v = v(x_t|x) - v_t^*(x_t)$. This discrepancy forces the network in the direction of $\Delta v \perp \nabla_x f_\theta$, leading to distortion of the flow map.

Intuitively, since the conditional velocity exhibits a larger variance than the marginal velocity, norm and variance of the corresponding gradient increase. This can lead training to diverge. Furthermore, this variance not only affects the gradient norm but also shifts the direction of training toward suboptimal flow maps. As it no longer guarantees injectivity and non-crossing trajectories, one of them

Table 2. **Consistency training results under different preconditioners.** *Multi-step FID* denotes the FIDs of pretrained networks for given ODE solver and sampling-step pairs. *Few-step FID* denotes two-step FIDs of consistency models initialized from the corresponding preconditioner (details in Appendix C.1).

Preconditioner	Multi-step FID↓	Few-step FID↓
Multi-step Baseline	1.21 (UCGM-S, 30-step)	2.52
LightningDiT	2.17 (Euler, 250-step)	9.59
In-house Flow Model	2.41 (UCGM-S, 30-step)	5.78
w/o Preconditioner	-	419.60 (Diverge)
Reported Baseline	1.21 (UCGM-S, 30-step)	1.42

can suffer from mode collapse or failure of locality-based editing.

Several studies [15, 31] have introduced network-induced couplings, which leverage neural networks to conduct the flow. They aim to reduce loss variance and provide a tighter bound to Eulerian distillation. However, a part of the gap $\mathcal{L}_{DT} - \mathcal{L}_{ED}$ remains, and thus convergence to the marginal velocity field is not guaranteed in the direct training setting (proof in Appendix A.8).

Instead of direct training, consistency training ensures the marginal flow map at its fixed point, even when guided by conditional velocity. However, it does not ensure convergence due to its gradient dynamics (proof in Appendix A.7).

Proposition 4. (*Instability of consistency training*) *Consistency training with a conditional velocity reduces to that with the marginal velocity under expectation.*

However, it lacks the curvature required to stabilize the optimum, ensuring only the existence of a fixed point that satisfies the Eulerian equation. Thus, the gradient dynamics may fail to converge.

This instability explains why FIDs degrade or oscillate after continued training: the model can escape the optimum and explore undesired fixed points due to the lack of curvature around the optimum.

Preconditioners. Despite various training techniques proposed in prior work, classical consistency training still suffers from reproducibility issues. In particular, recent consistency models rely on initialization with pretrained diffusion models. These pretrained models are often referred to as multi-step *preconditioners*, and we observe that the performance of consistency models varies depending on them.

As shown in Tab. 2, we evaluated an open-source continuous-time consistency model [38] on ImageNet-1K at 256×256 resolution. With the released multi-step baseline, the model achieved a result comparable to the reported one. However, when using the pretrained LightningDiT [41] or our in-house multi-step model, the FIDs were worse than the baseline. The resulting few-step FIDs are inconsistent: better multi-step preconditioners do not necessarily yield

better few-step models. We further observe that models diverge under random initialization, suggesting that the optimal flow map is inaccessible in this case. These highlight the sensitivity of consistency training to initialization and its inherent instability, making it difficult to reproduce.

Linearization Cost Hypothesis. Some studies [8, 9] enable training from scratch without a preconditioner. The key difference is that they allow $s < t$, while others fix $s = 0$. Intuitively, training long-range mappings is more challenging than short-range ones, since the linearization cost increases with step size. In both objectives, we observe that $s \rightarrow t$ amplifies the flow matching term, while $s \rightarrow 0$ amplifies a linearization term involving JVP. It is structurally more complex and may produce more undesired fixed points (Appendix A.9). We hypothesize that fixing $s = 0$ makes optimization more challenging, training less stable, while relaxing to $s < t$ balances the terms and mitigates instability.

4. Towards Reproducible Flow Map Training

From these observations, we identify two key factors for reproducibility: relaxation of s (linearization cost hypothesis), and marginal velocity guidance (Proposition 3). To make consistency training more reproducible, we relax the time condition to mitigate instability and leverage self-distillation to address suboptimality. Since prior work on self-distillation [3] relies on heuristics to stabilize training, we propose *improved Self-Distillation (iSD)*: (i) reducing reliance on heuristics and simplifying the training process by exploring the design space of flow maps, (ii) extending classifier-free guidance to self-distillation, achieving additional improvements.

4.1. Facilitating Consistency Training from Scratch

Relaxation of s . Based on the linearization cost hypothesis, we adopt relaxation of $s = 0$ to $s < t$, balancing the contributions of the flow matching and linearization terms. Instead of directly addressing the unstable gradient dynamics of consistency training, we leverage this relaxation to mitigate the instability indirectly. This approach still avoids the Hessian while empirically stabilizing optimization.

Marginal velocity guidance. Following Proposition 3, we consider marginal velocity guidance to address suboptimality. From the instantiations in Proposition 2, we identify that self-distillation follows the marginal velocity approximated by the network itself. Based on this, we train $F_\theta(x_t; t, t)$ via flow matching, while jointly optimizing Eulerian equation for $f_\theta(x_t; t, s)$ guided by its approximation:

$$\mathcal{L}_{SD} = \mathbb{E} [\|\partial_t f_\theta + F_{\theta-}(x_t; t, t) \cdot \nabla_x f_\theta\|_2^2]. \quad (12)$$

This objective encourages convergence to the marginal flow map, handling suboptimality of direct training (proof in Appendix B.1; see also [3]). As [25] shows, joint training

of flow matching can contribute to training stability from scratch, making it a natural choice in our case. Although consistency training already ensures the marginal flow map at its fixed point under conditional velocity guidance, incorporating self-distillation can further reduce loss variance and stabilize training (details in Appendix B.2).

4.2. Improved Self-Distillation

To incorporate various design choices and reduce the reliance on heuristics, we extend the self-distillation method using our generalized formulations and the explored design space. In the next section, we conduct ablation studies across these choices and present the best practice.

Reformulation. We leverage our generalized flow map defined in Eq. (6), and the objective can be expressed using the guidance velocity $v_\theta = F_\theta(x_t; t, t)$ as:

$$\mathcal{L}_{SD} = \mathbb{E} \left[\left\| A''_{t,s} x_t + A'_{t,s} (v_\theta - F_\theta) - A_{t,s} \frac{dF_\theta}{dt} \right\|_2^2 \right] \quad (13)$$

where $F_\theta = F_\theta(x_t; t, s)$. This extends the original self-distillation to arbitrary interpolations satisfying our assumptions. Next, we reformulate the objective, where the gradient remains identical while avoiding the Hessian:

$$\mathcal{L}_{SD-R} = \mathbb{E} [\|F_\theta - \text{sg}[F_{t_{\text{tgt}}}] \|_2^2], \quad (14)$$

$$F_{t_{\text{tgt}}} = F_\theta + A''_{t,s} x_t + A'_{t,s} (v_\theta - F_\theta) - A_{t,s} \frac{dF_\theta}{dt}. \quad (15)$$

JVP Approximation. We consider the approximation of JVP as follows to preserve the trajectory of marginal guidance. This approximation allows us to utilize Fully Sharded Data Parallel (FSDP), while requiring fewer computational resources and achieving better performance than direct computation.

$$\hat{x}_{t+\epsilon} = x_t + \epsilon \cdot v_\theta, \quad \hat{x}_{t-\epsilon} = x_t - \epsilon \cdot v_\theta \quad (16)$$

$$\frac{dF_\theta}{dt} \approx \frac{F_\theta(\hat{x}_{t+\epsilon}; t + \epsilon, s) - F_\theta(\hat{x}_{t-\epsilon}; t - \epsilon, s)}{2\epsilon} \quad (17)$$

Adaptive Weighting. Since the JVP exhibits instability and can cause training to diverge, several ideas have been proposed to stabilize the operation [3, 9, 23, 38]. Among these, we adopt adaptive weighting to preserve the intended guidance, as normalization and clipping may alter the guiding trajectory. To ensure stable joint training, we extend weighting to both objectives with $\eta = 0.01$ and $p = 1.0$.

$$\tilde{\mathcal{L}}_{t,s}(x_t, x) = \|v_\theta - v_t(x_t|x)\|_2^2 + \|F_\theta - \text{sg}[F_{t_{\text{tgt}}}] \|_2^2 \quad (18)$$

$$w_{t,s}(x_t, x) = (\text{sg}[\tilde{\mathcal{L}}_{t,s}(x_t, x)] + \eta)^{-p} \quad (19)$$

$$\mathcal{L}_{iSD} = \mathbb{E} [w_{t,s}(x_t, x) \cdot \tilde{\mathcal{L}}_{t,s}(x_t, x)] \quad (20)$$

From these settings, *training from scratch becomes stable, allowing us to eliminate additional heuristics introduced in prior work*, such as progressive distillation, annealing [3], small Fourier coefficients, double normalization, tangent warmup, and some regularizations [5, 23, 28]. Detailed training and sampling algorithms are provided in the Appendix, Algorithm 1 and Algorithm 2.

Classifier-free Guidance. Classifier-free Guidance (CFG) is an off-the-shelf method for boosting the performance of diffusion models. However, directly applying CFG to flow maps does not guarantee mappings along the CFG velocity field (details in Appendix B.4).

This motivates two extensions of the self-distillation: *Post-CFG* and *Pre-CFG*. Post-CFG operates as classical classifier-free guidance, applied at inference time:

$$\tilde{F}_c = F_\emptyset + \omega(F_c - F_\emptyset), \quad (21)$$

$$\tilde{f}_\theta(x_t; t, s, c) = \nu^{-1}(A'_{t,s}x_t - A_{t,s}\tilde{F}_c), \quad (22)$$

where $\emptyset$ is the null class label for unconditional generation, c is the conditional class label, and ω is the guidance scale. For brevity, we denote $F_\emptyset = F_\theta(x_t; t, s, \emptyset)$ and $F_c = F_\theta(x_t; t, s, c)$. Although this formulation is not guaranteed to follow the CFG velocity field, it can be easily applied after training. To ensure that the flow map follows the CFG field, we modify the flow matching objective:

$$\tilde{v}_t(x_t|x) = F_\emptyset + \omega(v_t(x_t|x) - F_\emptyset), \quad (23)$$

$$\tilde{\mathcal{L}}_{\text{CFM}} = \mathbb{E} \left[\|F_\theta(x_t; t, s, c) - \tilde{v}_t(x_t|x)\|_2^2 \right]. \quad (24)$$

Modified flow matching guarantees that F_θ converges to the ground-truth CFG velocity (proof in Appendix A.10). Note that the CFG velocity is Lipschitz under our assumption. Therefore, our propositions also hold for $\mathcal{L}_{\text{iSD}}$ when the flow matching objective is replaced accordingly, ensuring convergence to the flow map along the CFG velocity. We refer to this case as $\mathcal{L}_{\text{iSD-T}}$ (Training-time CFG). A detailed algorithm is provided in the Appendix, Algorithm 3.

5. Experiments

Experimental Settings. To evaluate our method, we conduct experiments on the ImageNet-1K [6] and CIFAR-10 [18] datasets. Following prior work, we use downsampled $32 \times 32 \times 4$ latent variables from 256×256 images encoded by a VAE [27], and adopt a DiT [24] architecture in BF16 precision. For CIFAR-10, we train the model in pixel space using UNet+ [35]. We evaluate both one-step and few-step generations, and sample quality is measured with FID [12] and Inception Score [30] (IS). Further implementation details are provided in Appendix C.2.

Table 3. **Effect of contributions.** 2-step evaluation results with standard deviations σ .

Case	FID $\downarrow$ ($\pm\sigma$)	IS $\uparrow$ ($\pm\sigma$)
Baseline ($\mathcal{L}_{\text{CT}}$, $s = 0$)	124.8 (± 2.32)	9.12 (± 0.37)
+ Joint training w/ $\mathcal{L}_{\text{CFM}}$	112.1 (± 10.1)	9.96 (± 2.90)
+ Relaxing to $s < t$	73.41 (± 1.30)	19.72 (± 0.50)
+ JVP Approximation	63.42 (± 0.17)	23.92 (± 0.11)
+ iSD-T ($\omega = 5.0$)	27.32 (± 0.35)	73.67 (± 1.88)
+ Trigonometric interp.	14.61 (± 0.05)	184.11 (± 0.50)
iSD-T ($\omega = 3.0$, Trig.)	15.63 (± 0.12)	135.51 (± 0.19)
MeanFlow ($\omega = 3.0$)	14.83 (± 2.00)	126.99 (± 24.4)

Table 4. **FIDs across design choices** (2-Step). The numbers in the header denote Post-CFG scales.

Loss	Interp.	JVP	1.0	3.0	5.0	7.0
$\mathcal{L}_{\text{CT}}$	Linear	Exact	73.41	39.33	53.38	68.82
$\mathcal{L}_{\text{CT}}$	Trig.	Exact	95.58	33.77	30.78	38.91
$\mathcal{L}_{\text{iSD}}$	Linear	Exact	115.60	76.26	79.86	89.88
$\mathcal{L}_{\text{iSD}}$	Trig.	Exact	107.32	46.48	41.92	48.24
$\mathcal{L}_{\text{CT}}$	Linear	Approx	63.42	33.53	50.17	66.68
$\mathcal{L}_{\text{iSD}}$	Linear	Approx	108.58	70.73	76.06	87.46
$\mathcal{L}_{\text{iSD-T}}$	Linear	Approx	27.32	45.80	58.43	66.54
$\mathcal{L}_{\text{iSD-T}}$	Trig.	Approx	14.61	20.37	29.56	40.32

5.1. Ablation Study

We conduct our ablation study on DiT-B/4, a base-sized diffusion transformer with 4×4 patch size. The model is trained for 400K steps with a batch size of 256. By default, we set $\omega = 5.0$ for $\mathcal{L}_{\text{iSD-T}}$ and use conditional velocity guidance for $\mathcal{L}_{\text{CT}}$. All model weights are randomly initialized.

Key factors. To validate our hypothesis, we conduct step-by-step experiments to make consistency training reproducible, as summarized in Tab. 3. We measure the variance of quantitative results across three runs with different random initializations. The reported results are median values.

We set the baseline as a continuous-time consistency model with linear interpolation and adaptive weighting. Applying joint training with flow matching and relaxing the time condition improves results and reduces variance, supporting the linearization cost hypothesis. We observe that self-distillation converges more slowly than the baseline (Tab. 4, Fig. 2a), while Pre-CFG $\mathcal{L}_{\text{iSD-T}}$ accelerates convergence and further improves performance. *With trigonometric interpolation, it achieves better results than MeanFlow*, which reported 15.53 FID on DiT-B/4 with training-time CFG $\omega = 3.0$, while we reproduced a 14.83 FID.

Interpolation and Post-CFG. As shown in Fig. 2a and Tab. 4, under conditional velocity guidance, linear interpo-

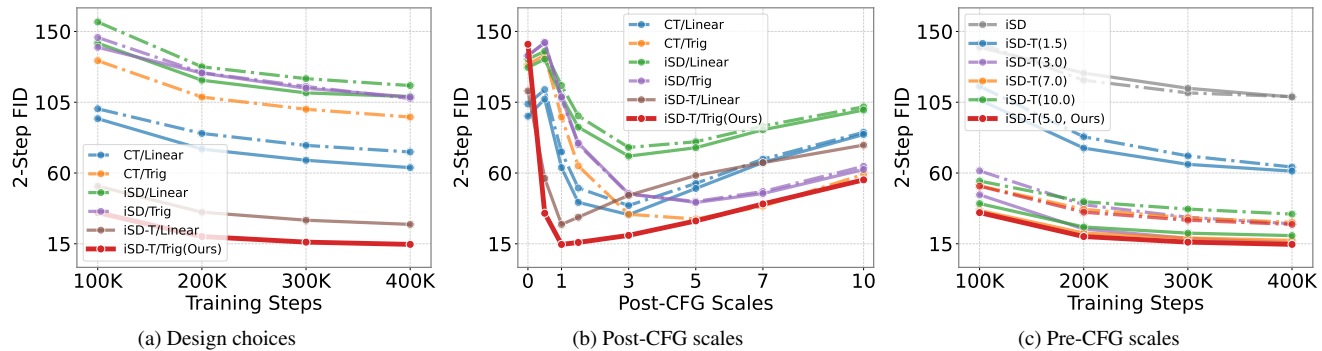


Figure 2. **Design choices.** (a) FIDs over training steps. Solid lines indicate the JVP approximation, and dash-dot lines indicate the direct JVP. (b) FIDs of Post-CFG over guidance scales. (c) FIDs of Pre-CFG over training steps. Solid lines indicate trigonometric interpolation, dash-dot lines indicate linear.

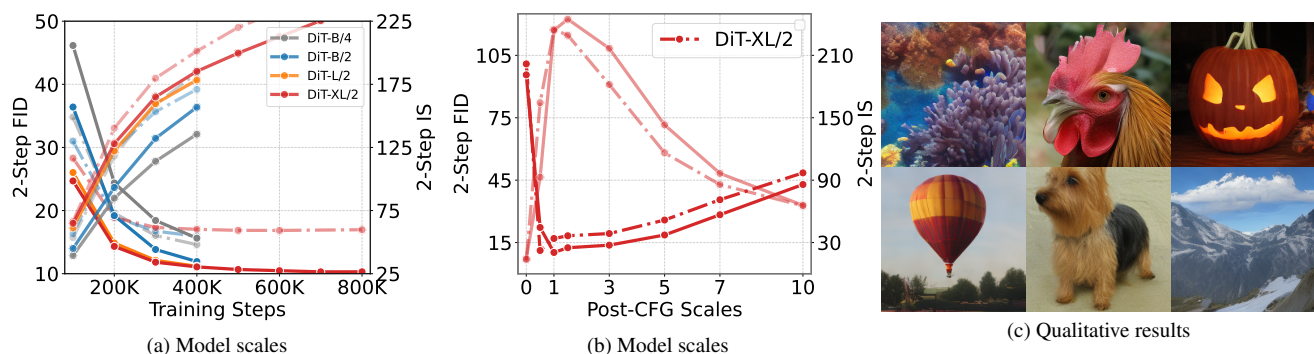


Figure 3. **Results of larger networks.** (a) Metrics across model scales over steps. Solid lines denote iSD-T with $\omega = 3.0$ and dash-dot lines denote $\omega = 5.0$. (b) Metrics over Post-CFG scales. (c) Two-step sampling results from the DiT-XL/2 trained with iSD-T at $\omega = 3.0$.

lation performs better than the trigonometric case. However, with self-distillation, the trigonometric interpolation achieves a lower FID and exhibits a larger performance gap under Post-CFG (Fig. 2b). Since the interpolation exhibits varying performance across different settings, further investigation into its properties is left for future work.

JVP operation. The JVP approximation consistently yields better results than direct computation, as illustrated in Fig. 2a. Under BF16 precision, it tends to be more robust, as direct computation is sensitive to numerical precision. Additionally, the VRAM requirement of the training process decreases from 25GB to 18GB on iSD-T, and the training speed improves from 1.48 steps/sec to 4.04 steps/sec on an A100 GPU. All subsequent experiments use the approximation by default.

Pre-CFG. As shown in Fig. 2c and Tab. 4, applying Pre-CFG $\mathcal{L}_{\text{iSD-T}}$ improves FIDs over vanilla $\mathcal{L}_{\text{iSD}}$. It also outperforms Post-CFG, which introduces a gap between the CFG trajectory and its own. Theoretically, Pre-CFG provides a more accurate approximation of the CFG field and demonstrates empirical improvements. Combining Post-CFG with

Pre-CFG often degrades performance, which can be interpreted as applying the guidance twice. When the resulting guidance scale exceeds the effective range, CFG may overly skew the trajectory, degrading performance.

We finalize the settings: (i) JVP approximation, (ii) trigonometric interpolation, and (iii) Pre-CFG.

Reproducibility. Following Tab. 3, we examine the variance of quantitative results to evaluate reproducibility. Our final setting outperforms MeanFlow on both metrics, while exhibiting lower standard deviations. These results indicate that iSD-T produces more reproducible outcomes.

Scalability. Fig. 3a presents the results across model scales. The FID improves as computation increases from DiT-B/4 to DiT-B/2 when $\omega = 3.0$. Further scaling from DiT-B/2 to DiT-XL/2 improves the FID, demonstrating consistent scaling behavior. However, with $\omega = 5.0$, the FID degrades as computation increases, suggesting that the effective guidance scale may vary across network sizes. For comparison with previous methods, we set $\omega = 3.0$.

Table 5. Comparison with prior work on CIFAR-10 and ImageNet-1K 256×256 .

Class-Conditional ImageNet-1K 256×256			Unconditional CIFAR-10		
METHOD	NFE ($\downarrow$)	FID ($\downarrow$)	METHOD	NFE ($\downarrow$)	FID ($\downarrow$)
Consistency Training Methods			Distillation-based Methods		
iCT [32]	2	20.3	2-Rectified Flow [21]	1	4.85
Shortcut Model [8]	1	10.6	DMD [43]	1	3.77
	4	7.8	Consistency Training Methods		
UCGM [38] (SD-VAE)	1	2.10	iCT [32]	1	2.83
IMM [45]	1 $\times$ 2	7.77	sCT [23]	1	2.97
	8 $\times$ 2	1.99	UCGM [38]	1	2.82
MeanFlow [9]	1	3.43	MeanFlow [9]	1	2.92
	2	2.20	Self-distillation [3]	1	14.13
Ours			Ours		
iSD-T ($\omega = 3.0$)	1	11.74	iSD	1	3.64
	2	9.70			

5.2. Comparison with Prior Work

In Tab. 5, we compare our work with previous methods on CIFAR-10. We train UNet+ in a class-unconditional setting and therefore use vanilla iSD without CFG to ensure a fair comparison. This setup achieves improved results over the original self-distillation, with performance comparable to recent approaches. Compared to the original self-distillation, we find that the trigonometric interpolation and the JVP approximation contribute to improving the result, as demonstrated in our ablation studies. Additionally, adaptive weighting sufficiently stabilizes training while reducing the heuristics introduced in self-distillation, such as the progressive distillation scheme and the auxiliary MLP network for loss weighting.

We train DiT-XL/2 with iSD-T on ImageNet 256×256 for 800K steps following prior work. Our model achieves results on par with our reproduced consistency model (Tab. 2, FID 10.01), iCT, and Shortcut Model, though it does not match the state-of-the-art methods. Since our method requires neither additional networks nor pretrained models, the training process is simplified. Additionally, its reproducibility has been validated in the previous section. However, we observe that iSD converges more slowly than consistency training, as it first learns the marginal velocity and then the flow map guided by itself. Additionally, network scaling does not improve performance as expected, iSD-T outperforms MeanFlow on DiT-B/4 but underperforms on DiT-XL/2. This can be viewed as a trade-off between reproducibility and performance, and achieving scalable yet reproducible training remains as future work.

6. Conclusion

We introduced a generalized flow map framework that unifies recent consistency-based generative models under

the Eulerian equation. This highlights the suboptimality and instability of existing consistency-based approaches, explaining their limited reproducibility. To address these issues, we propose *improved Self-Distillation*, which reduces reliance on heuristics and simplifies the training process. We further extend it to classifier-free guidance for flow maps, achieving additional performance gains. Empirically, our method achieves reproducible training on ImageNet-1K. These results establish a theoretical and empirical foundation for reproducible consistency training.

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Reproducible Consistency Training via Improved Self-Distillation Guidance

Supplementary Material (Appendix)

A. Theoretical Analysis of Flow Map Models

A.1. Mean collapse of Diffusion and Flow Matching Models

Posterior Distribution. First, consider the data distribution $p_X(x) = \mathcal{N}(\mu_X, \sigma_X^2)$ and the interpolation $x_t = \alpha_t x + \sigma_t z$. The conditional distribution is given by $p(x_t = y|x) = \mathcal{N}(y; \alpha_t x, \sigma_t^2)$. By Bayes' rule,

$$\begin{aligned} p(x|x_t = y) &\propto p(x_t = y|x)p(x) \\ &= \mathcal{N}(y; \alpha_t x, \sigma_t^2) \mathcal{N}(x; \mu_X, \sigma_X^2) \\ &\propto \exp\left(-\frac{(y - \alpha_t x)^2}{2\sigma_t^2} - \frac{(x - \mu_X)^2}{2\sigma_X^2}\right) \\ &= \exp\left(\left(\frac{1}{2\sigma_X^2} + \frac{\alpha_t^2}{2\sigma_t^2}\right)x^2 - \left(\frac{\mu_X}{\sigma_X^2} + \frac{\alpha_t y}{\sigma_t^2}\right)x + \left(\frac{\mu^2}{2\sigma_X^2} + \frac{y^2}{2\sigma_t^2}\right)\right) \end{aligned}$$

This can be organized as a Gaussian with a closed form $p(x|x_t = y) = \mathcal{N}(\mu_{x|y,t}, \sigma_{x|y,t}^2)$ where

$$\mu_{x|y,t} = \frac{\alpha_t \sigma_X^2 y + \mu_X \sigma_t^2}{\sigma_t^2 + \sigma_X^2 \alpha_t^2}, \quad \sigma_{x|y,t}^2 = \frac{\sigma_X^2 \sigma_t^2}{\sigma_t^2 + \sigma_X^2 \alpha_t^2}$$

Extending the data distribution to a mixture of Gaussians $p_X(x) = \sum_i \pi_i \mathcal{N}(x; \mu_i, \sigma_i^2)$, we introduce the latent variable π for handling π_i :

$$p(\pi = i) = \pi_i, \quad p(x|\pi = i) = \mathcal{N}(x; \mu_i, \sigma_i^2)$$

Then, the marginal distribution $p(x_t = y|\pi = i)$ can be expressed as

$$\begin{aligned} p(x_t = y|\pi = i) &= \int p(x_t = y|x)p(x|\pi = i)dx \\ &= \int \mathcal{N}(y; \alpha_t x, \sigma_t^2) \mathcal{N}(x; \mu_i, \sigma_i^2) dx \\ &= \mathcal{N}(y; \alpha_t \mu_i, \alpha_t^2 \sigma_i^2 + \sigma_t^2) \end{aligned}$$

And we define responsibilities $r_i(y)$ as posterior distribution

$$r_{i,t}(y) = p(\pi = i|x_t = y) = \frac{p(x_t = y|\pi = i)p(\pi = i)}{\sum_j p(x_t = y|\pi = j)p(\pi = j)} = \frac{\pi_i \mathcal{N}(y; \alpha_t \mu_i, \alpha_t^2 \sigma_i^2 + \sigma_t^2)}{\sum_j \pi_j \mathcal{N}(y; \alpha_t \mu_j, \alpha_t^2 \sigma_j^2 + \sigma_t^2)}$$

Therefore, the posterior distribution $p(x|x_t = y)$ is

$$\begin{aligned} p(x|x_t = y) &= \sum_i p(\pi = i|x_t = y)p(x|x_t = y, \pi = i) \\ &= \sum_i r_{i,t}(y) \mathcal{N}(y; \alpha_t x, \sigma_t^2) \mathcal{N}(x; \mu_i, \sigma_i^2) \\ &= \sum_i r_{i,t}(y) \mathcal{N}(x; \mu_{x|i,y,t}, \sigma_{x|i,y,t}^2) \\ &\text{where } \mu_{x|i,y,t} = \frac{\alpha_t \sigma_i^2 y + \mu_i \sigma_t^2}{\sigma_t^2 + \sigma_i^2 \alpha_t^2}, \quad \sigma_{x|i,y,t}^2 = \frac{\sigma_i^2 \sigma_t^2}{\sigma_t^2 + \sigma_i^2 \alpha_t^2} \end{aligned}$$

Particularly, we observe that $\mu_{x|i,y,1} = \mu_i$, $\sigma_{x|i,y,1}^2 = \sigma_i^2$ and $r_{i,1}(y) = \pi_i$.

One-step Generation. Under the linear trajectory $x_t = (1 - t)x + tz$, the conditional velocity is $v_t(x_t|x) = z - x$. Thus, one-step generation is defined by

$$\begin{aligned} f_F(x_t; t) &= x_t - tF^*(x_t; t) = x_t - t\mathbb{E}_{x|x_t}[v_t(x_t|x)] \\ &= \mathbb{E}_{x,z|x_t}[x_t - t(z - x)] \\ &= \mathbb{E}_{x|x_t}[x] \end{aligned}$$

In the unimodal Gaussian case, $\mathbb{E}_{x|z}[x] = \mu_{x|z,1} = \mu_X$ and the one-step generated samples collapse to the mean of the data distribution. Similarly, in the mixture of Gaussians case, one-step generated samples collapse to the mixture mean.

$$f_F(z) = \mathbb{E}_{x|z}[x] = \sum_i r_i(z) \mu_{i|z,1} = \sum_i \pi_i \mu_i = \mu_X$$

Thus, one-step generation collapses to the data mean μ_X regardless of the input. $\square$

A.2. Injectivity of Flow Map

Since the marginal velocity is assumed to be Lipschitz continuous, the Picard-Lindelöf theorem guarantees a unique solution to the ODE $dx_t = v_t^*(x_t)dt$ for any initial value. Non-crossing trajectory follows directly, since any crossing would contradict the uniqueness. Thus, since the flow map is formulated as the solution of the ODE with the initial value x_t , it is well-defined and the solution x_s is uniquely determined by non-crossing, ensuring the injectivity of the flow map.

A.3. Eulerian Equation and Uniqueness of Flow Map

Suppose the ground-truth flow map is defined as

$$f_{t,s}^*(x_t) = x_t + \int_t^s v_\tau^*(x_\tau) d\tau = x_s$$

By construction, the identity mapping $f_{t,s}^*(f_{s,t}^*(x_s)) = x_s$ satisfies. Differentiating both sides w.r.t. t yields

$$\frac{d}{dt} f_{t,s}^*(f_{s,t}^*(x_s)) = \partial_t f_{t,s}^*(f_{s,t}^*(x_s)) + \partial_t f_{s,t}^*(x_s) \cdot \nabla_x f_{t,s}^*(f_{s,t}^*(x_s)) = \frac{d}{dt} x_s = 0$$

Using $f_{s,t}^*(x_s) = x_t$ and $\partial_t f_{s,t}^*(x_s) = \partial_t x_t = v_t^*(x_t)$, we obtain the Eulerian equation:

$$\frac{d}{dt} f_{t,s}^*(x_t) = \partial_t f_{t,s}^*(x_t) + v_t^*(x_t) \cdot \nabla_x f_{t,s}^*(x_t) = 0$$

Suppose a trainable network $f_\theta(x; t, s) = f_{t,s}^\theta(x)$ is continuous in x, t, s , Lipschitz continuous in x , and satisfies the boundary condition $f_{s,s}^\theta(x) = x$ for all s . If f_θ satisfies the Eulerian equation, $f_{t,s}^\theta$ remains constant along the characteristic curve induced by $v_t^*(x_t)$.

Let χ_τ denotes the characteristic curve defined on $[s, t]$ by $\chi_t = x$ and $\chi'_\tau = v_\tau^*(\chi_\tau)$. Along this curve, $f_{\tau,s}^\theta$ is constant and evaluating at $\tau = t$ and $\tau = s$ yields

$$f_{t,s}^\theta(x) = f_{t,s}^\theta(\chi_t) = f_{s,s}^\theta(\chi_s) = \chi_s = f_{t,s}^*(x)$$

since $f_{t,s}^*$ generates the characteristic curve χ_τ by its definition. Thus, the learned mapping coincides with the exact flow map. $\square$

A.4. Interpolation condition for guaranteeing the convergence

We begin by explicitly deriving the solution of the Eulerian equation. For $f_{t,s}^\theta(x_t) = \nu_t^{-1}(A'_{t,s}x_t - A_{t,s}F_{t,s}^\theta(x_t))$ in Eq. (6), let $n = A'_{t,s}x_t - A_{t,s}F_{t,s}^\theta(x_t)$, which simplifies to $n = A'x_t - AF$ and $f = \nu_t^{-1}n$. Then, differentiation with respect to timestep t yields

$$\begin{aligned} n' &= A''x_t + A'(v_t - F) - AF' \\ f' &= [A''x_t + A'(v_t - F) - AF'] \frac{\nu_t}{\nu_t^2} + [A'x_t - AF] \frac{\nu_t'}{\nu_t^2} \end{aligned}$$

Assuming $f' = 0$ and $\nu_t \neq 0$, we obtain

$$A\nu_t F' = -(A'\nu_t + A\nu_t')F + [A''\nu_t x_t + A'\nu_t v_t + A'\nu_t' x_t]$$

which further simplifies to

$$A\nu_t F' = -[A\nu_t]'F + [A\nu_t x_t]' = BF' = -B'F + D'$$

where $B = A\nu_t$, $D = A'\nu_t x_t$. Since $BF' + B'F = [BF]'$, we obtain $[BF]' = D'$, which yields $A\nu_t F = A'\nu_t x_t + C$ for some integration constant C . Assuming $A = A_{t,s} = \alpha_s \sigma_t - \sigma_s \alpha_t \neq 0$ for $t \neq s$, this gives $F = A'A^{-1}x_t + C(A\nu_t)^{-1}$, and hence

$$f_{t,s}^*(x_t) = \nu_t^{-1}[A'x_t - A'x_t - C\nu_t^{-1}] = -C\nu_t^{-2}$$

If ν_t is a time-dependent scalar, the global optimum $f^* = -C'\nu_t^{-2}$ is itself time-dependent, so the Eulerian equation can't vanish. If $C = 0$, the solution collapses to the trivial case $f_{t,s}^*(x_t) = 0$. Therefore, ν_t must be a time-independent constant. $\square$

Additional Observation. Suppose that the monotonically increasing γ_t over $t \in [0, 1]$ satisfying the boundary conditions $\gamma_1 = 1$ and $\gamma_0 = 0$. Consider the interpolation defined by $\alpha_t = (1 - \gamma_t)^c$ and $\sigma_t = \gamma_t^c$ for some constant $c \in [0.5, 1]$. Then, ν_t can be written as $\nu_t = c(1 - \gamma_t)^{c-1} \gamma_t^{c-1} \gamma_t'$. Imposing $\nu_t = \nu$ gives $\gamma_t' = \nu[c(1 - \gamma_t)^{c-1} \gamma_t^{c-1}]^{-1}$ and $c(1 - \gamma)^{c-1} \gamma^{c-1} d\gamma = \nu dt$. Integrating both sides yields

$$c \int (1 - \gamma)^{c-1} \gamma^{c-1} d\gamma = \nu \int dt = \nu(t + C)$$

where the constant C vanishes due to $\gamma_0 = 0$. For the incomplete beta function B , this becomes

$$cB(\gamma_t; c, c) = c \int_0^{\gamma_t} (1 - \eta)^{c-1} \eta^{c-1} d\eta = \nu \int_0^t d\tau = \nu t$$

By the boundary condition, $cB(c, c) = cB(1; c, c) = \nu$, and thus γ_t is characterized by

$$cB(\gamma_t; c, c) = \nu \frac{cB(\gamma_t; c, c)}{cB(c, c)} = \nu I_{\gamma_t}(c, c) = \nu t \implies \gamma_t = I_t^{-1}(c, c)$$

where I denotes the regularized incomplete beta function.

In this case, γ_t is characterized regardless of ν . Particularly, when $c = 0.5$, we obtain $\gamma_t = \sin^2(\frac{\pi}{2}t)$, which yields trigonometric interpolation $\alpha_t = \cos(\frac{\pi}{2}t)$ and $\sigma_t = \sin(\frac{\pi}{2}t)$. On the other hand, when $c = 1$, we have $\gamma_t = t$, $\alpha_t = 1 - t$, $\sigma_t = t$, and which reduces to linear interpolation. Interpolating c between 0.5 and 1.0 provides a promising design choice for formulating the consistency model.

A.5. Recent Consistency-based Generative Models are Flow Map Models

sCT. Under trigonometric interpolation $x_t = \cos(t)x + \sin(t)z$,

$$f_\theta(x_t; t, s) = \cos(s - t)x_t + \sin(s - t)F_\theta(x_t; t, s)$$

When $s = 0$

$$f_\theta(x_t; t) = \cos(t)x_t - \sin(t)F_\theta(x_t; t)$$

which exactly recovers the sCT formulation. If consistency training is formulated without the stop-gradient operation, then the objective reduces to the direct training objective as $\Delta t \rightarrow 0$.

$$\begin{aligned} & \mathbb{E} \left[\|f_\theta(x_t; t, s) - f_\theta(x_{t-\Delta t}; t - \Delta t, s)\|_2^2 \right] \\ &= \mathbb{E} \left[\left\| f_\theta(x_t; t, s) - [f_\theta(x_t; t, s) - \partial_t f_\theta(x_t; t, s) \cdot \Delta t - \nabla_x f_\theta(x_t; t, s) \cdot v(x_t|x) \cdot \Delta t + O(\Delta t^2)] \right\|_2^2 \right] \\ &= \Delta t^2 \cdot \mathbb{E} \left[\left\| \partial_t f_\theta(x_t; t, s) + v(x_t|x) \cdot \nabla_x f_\theta(x_t; t, s) \right\|_2^2 \right] + O(\Delta t^3) \end{aligned}$$

However, if we utilize the stop-gradient, the continuous-time consistency training objective is defined as

$$\begin{aligned}
& \nabla_{\theta} \mathbb{E} \left[\|f_{\theta}(x_t; t, s) - f_{\theta^-}(x_{t-\Delta t}; t - \Delta t, s)\|_2^2 \right] \\
&= \mathbb{E} \left[2 \nabla_{\theta} f_{\theta}(x_t; t, s) \cdot (f_{\theta^-}(x_t; t, s) - f_{\theta^-}(x_{t-\Delta t}; t - \Delta t, s)) \right] \\
&= 2 \Delta t \cdot \nabla_{\theta} \mathbb{E} \left[f_{\theta}(x_t; t, s) \cdot \frac{f_{\theta^-}(x_t; t, s) - f_{\theta^-}(x_{t-\Delta t}; t - \Delta t, s)}{\Delta t} \right] \\
&\implies \mathcal{L}_{CT} = \mathbb{E} \left[f_{\theta}(x_t; t, s) \frac{df_{\theta^-}(x_t; t, s)}{dt} \right]
\end{aligned}$$

By the mean value theorem,

$$f_{\theta^-}(x_t; t, s) = f_{\theta}(x_t; t, s) + \nabla_{\theta} f_{\zeta}(x_t; t, s) \cdot (\theta^- - \theta)$$

for the parameter ζ lying between θ and θ^- . Using this, we can show that

$$\begin{aligned}
& \mathbb{E} \left[\|f_{\theta}(x_t; t, s) - f_{\theta^-}(x_{\hat{t}}; \hat{t}, s)\|_2^2 \right] \\
&= \mathbb{E} \left[\|f_{\theta}(x_t; t, s) - f_{\theta}(x_{\hat{t}}; \hat{t}, s) - \nabla_{\theta} f_{\zeta}(x_{\hat{t}}; \hat{t}, s) \cdot (\theta^- - \theta)\|_2^2 \right] \\
&= \mathbb{E} \left[\|\partial_t f_{\theta}(x_t; t, s) \cdot \Delta t + \nabla_x f_{\theta}(x_t; t, s) \cdot v(x_t|x) \cdot \Delta t - \nabla_{\theta} f_{\zeta}(x_{\hat{t}}; \hat{t}, s) \cdot (\theta^- - \theta) + O(\Delta t^2)\|_2^2 \right] \\
&= \mathcal{L}_{DT} \cdot \Delta t^2 - 2 \mathbb{E} \left[L f_{\theta}(x_t; t, s) \cdot J_{\hat{t}, s}^{\zeta} \cdot \Delta \theta \right] \cdot \Delta t + \mathbb{E} \left[\|J_{\hat{t}, s}^{\zeta} \cdot \Delta \theta\|_2^2 \right] + O(\Delta t^3) + O(\Delta t^2 \|\Delta \theta\|)
\end{aligned}$$

where $\hat{t} = t - \Delta t$, $\Delta \theta = \theta^- - \theta$, $J_{\hat{t}, s}^{\zeta} = \nabla_{\theta} f_{\zeta}(x_{\hat{t}}; \hat{t}, s)$, and the operator is defined as $Lf = \partial_t f + v \cdot \nabla_x f$. In this case, since f is Lipschitz and has a bounded first derivative, Lf and J are bounded. Hence, $2(Lf \cdot \Delta t - J \cdot \Delta \theta) \cdot O(\Delta t^2) = O(\Delta t^3) + O(\Delta t^2 \|\Delta \theta\|)$. If we set $\theta^- = \text{sg}[\theta]$, this reduces to

$$\mathbb{E} \left[\|f_{\theta}(x_t; t, s) - f_{\theta^-}(x_{t-\Delta t}; t - \Delta t, s)\|_2^2 \right] = \mathcal{L}_{DT} \cdot \Delta t^2 + O(\Delta t^3)$$

Thus, the formulation can be interpreted as the direct training objective.

From another perspective, since the time derivative of $f_{t,s}^{\theta}(x_t) = f_{\theta}(x_t; t, s)$ is given by

$$\frac{df_{t,s}^{\theta}(x_t)}{dt} = \partial_t f_{t,s}^{\theta}(x_t) + v(x_t|x) \cdot \nabla_x f_{t,s}^{\theta}(x_t) = (L_* f_{t,s}^{\theta})(x_t) + \Delta v \cdot \nabla_x f_{t,s}^{\theta}(x_t)$$

where $\Delta v = v_t(x_t|x) - v_t^*(x_t)$ and $L_* f_{t,s} = \partial_t f_{t,s} + v_t^* \cdot \nabla_x f_{t,s}$, the objective can be written as

$$\mathcal{L}_{CT} = \mathbb{E} \left[f_{t,s}^{\theta} \left(L_* f_{t,s}^{\theta^-} \right) + f_{t,s}^{\theta} \left(\Delta v \cdot \nabla_x f_{t,s}^{\theta^-} \right) \right]$$

The first term on the right-hand side corresponds to Eulerian distillation. By the tower property, the second term vanishes under conditional expectation:

$$\begin{aligned}
\mathbb{E}_{x,z,t,s} \left[f_{t,s}^{\theta} (\Delta v \cdot \nabla_x f_{t,s}^{\theta^-}) \right] &= \mathbb{E}_{x,z,t,s} \left[\mathbb{E}_{\tilde{x}|x_t} \left[f_{t,s}^{\theta} (\Delta v \cdot \nabla_x f_{t,s}^{\theta^-}) \right] \right] \\
&= \mathbb{E}_{x,z,t,s} \left[f_{t,s}^{\theta} \left(\mathbb{E}_{\tilde{x}|x_t} [\Delta v] \cdot \nabla_x f_{t,s}^{\theta^-} \right) \right] \\
&= \mathbb{E}_{x,z,t,s} \left[f_{t,s}^{\theta} \cdot 0 \cdot \nabla_x f_{t,s}^{\theta^-} \right] = 0
\end{aligned}$$

Thus, $\mathcal{L}_{CT}$ reduces in value to Eulerian distillation even along the conditional trajectory. However, the gradients of Eulerian distillation and consistency training differ, and their training dynamics may therefore exhibit distinct behaviors. The instability of these gradient dynamics is discussed in Appendix A.7. $\square$

MeanFlow. Suppose a flow map model under linear interpolation.

$$f_{\theta}(x_t; t, s) = x_t + (s - t)F_{\theta}(x_t; t, s), \quad x_t = (1 - t)x + tz$$

The corresponding direct training objective is

$$\mathcal{L}(\theta) = \mathbb{E} \left[\left\| \partial_t f_{t,s}^\theta(x_t) + v(x_t|x) \cdot \nabla_x f_{t,s}^\theta(x_t) \right\|_2^2 \right] = \mathbb{E} \left[\left\| \frac{df_{t,s}^\theta(x_t)}{dt} \right\|_2^2 \right]$$

where

$$\frac{d}{dt} f_\theta(x_t; t, s) = v_t - F_\theta(x_t; t, s) + (s - t) \frac{d}{dt} F_\theta(x_t; t, s)$$

Recall the MeanFlow objective from [9]

$$\mathcal{L}(\theta) = \mathbb{E} [\|u_\theta(z_t; r, t) - \text{sg}[v_t - (t - r)(v_t \cdot \partial_z u_\theta + \partial_t u_\theta)]\|_2^2]$$

Rewrite the MeanFlow objective by using flow map notation and transform

$$\begin{aligned} & \nabla_\theta \mathbb{E} [\|F_\theta(x_t; t, s) - \text{sg}[v_t(x_t|x) - (t - s)(v_t(x_t|x) \cdot \nabla_x F_\theta + \partial_t F_\theta)]\|_2^2] \\ &= \nabla_\theta \mathbb{E} \left[\left\| v_t(x_t|x) - F_\theta(x_t; t, s) + (s - t) \cdot \frac{d}{dt} F_{\theta-}(x_t; t, s) \right\|_2^2 \right] \\ &= \nabla_\theta \mathbb{E} \left[\left\| F_\theta(x_t; t, s) - F_{\theta-}(x_t; t, s) - \left[v_t(x_t|x) - F_{\theta-}(x_t; t, s) + (s - t) \frac{d}{dt} F_{\theta-}(x_t; t, s) \right] \right\|_2^2 \right] \\ &= \nabla_\theta \mathbb{E} \left[\frac{1}{t - s} f_\theta(x_t; t, s) \frac{df_{\theta-}(x_t; t, s)}{dt} \right] \end{aligned}$$

Thus, the MeanFlow objective is a special case of the continuous-time consistency training with conditional velocity under linear interpolation. $\square$

Shortcut Model. From [8], the Shortcut Model objective consists of the flow matching objective and the consistency objective.

$$\mathcal{L}(\theta) = \mathbb{E} [\|s_\theta(x_t; t, 0) - v_t\|_2^2 + \|s_\theta(x_t; t, 2d) - [s_\theta(x_t; t, d) + s_\theta(x'_{t+d}; t + d, d)]/2\|_2^2]$$

with $x'_{t+d} = x_t + s_\theta(x_t, t, d)$. By setting $d = s - t$ and $F_\theta(x_t; t, s) = s_\theta(x_t; t, s - t)$, sampling $t \sim \mathcal{U}[0, 1]$, and choosing $s = t - 2^{-d'}$ for $d' \sim \text{Cat}[1, 7]$, we obtain the flow map under linear interpolation

$$f_\theta(x_t; t, s) = x_t + (s - t)F_\theta(x_t; t, s) = x'_{t+d}$$

We can rewrite the flow matching objective of the Shortcut model as

$$\|s_\theta(x_t; t, 0) - v_t\|_2^2 = \|F_\theta(x_t; t, t) - v_t(x_t|x)\|_2^2$$

For $r = s + d$, the consistency objective of the Shortcut Model can be written in the form of the CTM(Consistency Trajectory Model)

$$\begin{aligned} & \|s_\theta(x_t; t, 2d) - [s_\theta(x_t; t, d) + s_\theta(x'_{t+d}; t + d, d)]/2\|_2^2 \\ &= \|F_\theta(x_t; t, r) - [F_\theta(x_t; t, s) + F_\theta(f_\theta(x_t; t, s); s, r)]/2\|_2^2 \\ &= \frac{1}{4d^2} \|x_t + 2d \cdot F_\theta(x_t; t, r) - x_t - 2d[F_\theta(x_t; t, s) + F_\theta(f_\theta(x_t; t, s); s, r)]/2\|_2^2 \\ &= \frac{1}{4d^2} \|x_t + 2d \cdot F_\theta(x_t; t, r) - [x_t + d \cdot F_\theta(x_t; t, s) + d \cdot F_\theta(f_\theta(x_t; t, s); s, r)]\|_2^2 \\ &= \frac{1}{4d^2} \|f_\theta(x_t; t, r) - [f_\theta(x_t; t, s) + d \cdot F_\theta(f_\theta(x_t; t, s); s, r)]\|_2^2 \\ &= \frac{1}{4d^2} \|f_\theta(x_t; t, r) - f_\theta(f_\theta(x_t; t, s); s, r)\|_2^2 \end{aligned}$$

Hence, the objective of the Shortcut Model is

$$\mathcal{L}(\theta) = \mathbb{E} \left[\|F_\theta(x_t; t, t) - v_t\|_2^2 + \frac{1}{4d^2} \|f_\theta(x_t; t, r) - f_\theta(f_\theta(x_t; t, s); s, r)\|_2^2 \right]$$

With the Taylor approximation of $F_{t,s} = F_{t,s}(x_t) = F_\theta(x_t; t, s)$

$$F_{t,r} = F_{t,s} + d \cdot \partial_s F_{t,s} + O(d^2), \quad F_{s,r} = F_{t,s} + d \cdot \partial_t F_{t,s} + d \cdot \partial_s F_{t,s} + d \cdot F_{t,s}^T \nabla_x F_{t,s} + O(d^2)$$

we obtain

$$\begin{aligned} & d[2F_{t,r} - F_{t,s} - F_{s,r}] \\ &= d[2[F_{t,s} + d \cdot \partial_s F_{t,s}] - F_{t,s} - [F_{t,s} + d \cdot \partial_t F_{t,s} + d \cdot \partial_s F_{t,s} + d \cdot F_{t,s}^T \nabla_x F_{t,s}]] + O(d^3) \\ &= d^2[\partial_s F_{t,s} - \partial_t F_{t,s} - F_{t,s}^T \nabla_x F_{t,s}] + O(d^3) \end{aligned}$$

Thus,

$$\frac{1}{4d^2} \|f_{t,r}(x_t) - f_{s,r}(f_{t,s}(x_t))\|_2^2 = \frac{d^2}{4} \|\partial_s F_{t,s} - \partial_t F_{t,s} - F_{t,s}^T \nabla_x F_{t,s}\|_2^2 + O(d^3)$$

The differentiation of the linear flow map with respect to timestep t is given by

$$f'_{t,s}(x_t) = v_t^* - F_{t,s}(x_t) + (s - t) \cdot (\partial_t F_{t,s} + v_t^* \cdot \nabla_x F_{t,s})$$

With the Taylor approximation and the relation $F_{t,t} \approx v_t^*$ obtained from $\mathcal{L}_{\text{CFM}}$, we have

$$F_{t,s} = F_{t,t} + d \cdot \partial_s F_{t,t} + O(d^2) \approx v_t^* + d \cdot \partial_s F_{t,t} + O(d^2)$$

The identity $\partial_s F_{t,s} = \partial_s F_{t,t} + O(d)$ implies

$$v_t^* - F_{t,s}(x_t) = -d \cdot \partial_s F_{t,s} + O(d^2)$$

Hence,

$$f'(x_t) = d[\partial_t F_{t,s} + v_t^* \cdot \nabla_x F_{t,s} - \partial_s F_{t,s}] + O(d^2)$$

Since $v_t^* \approx F_{t,t} = F_{t,s} + O(d)$, we further obtain

$$\begin{aligned} & f'(x_t) = d[\partial_t F_{t,s} + F_{t,s} \cdot \nabla_x F_{t,s} - \partial_s F_{t,s}] + O(d^2) \\ \implies \mathcal{L} &= \mathbb{E}_{x,z,t,s} \left[\|F_\theta(x_t; t, t) - v_t\|_2^2 + \frac{1}{4} \left\| \frac{d}{dt} f_\theta(x_t; t, s) \right\|_2^2 + O(d^3) \right] \end{aligned}$$

We observe that there is a discrepancy between Eulerian distillation, expressed as $\|v_t - F_{t,s} + d(\partial_t F_{t,s} + v_t^* \cdot \nabla_x F_{t,s})\|_2^2$. The first v_t^* term corresponds to $F_{t,t}$, while the second v_t corresponds to $F_{t,s}$. Alternatively, since $F_{t,t} = F_{t,s} + O(d)$, both terms can be represented in terms of $F_{t,t}$.

In the case of $F_{t,t}$, the model learns $F_{t,t}^\theta(x_t) \approx v_t^*(x_t)$ due to the loss term of $\|F_{t,t}^\theta(x_t) - v_t(x_t|x)\|_2^2$. This can be interpreted as the model learning a flow map corresponding to the trajectory induced by an approximated marginal velocity. Therefore, the Shortcut Model can be seen as Eulerian distillation under an $O((s - t)^3)$ bound. Since Flow Map Models typically operate under the assumption $s \in [0, t]$, the Shortcut model's sampling scheme with $d \in [2^{-7}, 1]$ makes this error non-negligible. $\square$

Consistency Flow Matching. For linear interpolation $x_t = (1 - t)x + tz$, define

$$f_\theta(x_t; t, s = 0) = x_t - tF_\theta(x_t; t, s = 0) \implies f_\theta(x_t; t) = x_t - tF_\theta(x_t; t)$$

Then, the Consistency Flow Matching objective from [40] becomes

$$\mathcal{L}(\theta) = \mathbb{E} [\|f_\theta(x_t; t) - f_\theta(x_{t-\Delta t}; t - \Delta t)\|_2^2 + \alpha \|F_\theta(x_t; t) - F_\theta(x_{t-\Delta t}; x - \Delta t)\|_2^2]$$

The first term on the right side is the Taylor approximation of the consistency training objective, and the second term is the regularizer. Hence, we can interpret Consistency Flow Matching as a training flow map model via the approximation with regularization. $\square$

UCGM. For arbitrary interpolation of α_t, σ_t , setting $s = 0$ yields

$$f_\theta(x_t; t) = \nu_t^{-1}(\sigma'_t x_t - \sigma_t F_\theta)$$

We can reformulate the objective while keeping the gradient unchanged:

$$\begin{aligned} & \nabla_\theta \|f_\theta(x_t; t) - f_{\theta-}(x_{\lambda t}; \lambda t)\|_2^2 \\ &= 2[\nabla_\theta f_\theta(x_t; t)]^T (t - \lambda t) \frac{f_{\theta-}(x_t; t) - f_{\theta-}(x_{\lambda t}; \lambda t)}{t - \lambda t} \\ &\propto [\nabla_\theta f_\theta(x_t; t)]^T \frac{f_{\theta-}(x_t; t) - f_{\theta-}(x_{\lambda t}; \lambda t)}{t - \lambda t} \\ &= \frac{\sigma_t}{\nu_t} [\nabla_\theta F_\theta(x_t; t)]^T \frac{f_{\theta-}(x_t; t) - f_{\theta-}(x_{\lambda t}; \lambda t)}{t - \lambda t} \\ &= \nabla_\theta \left\| F_\theta(x_t; t) - F_{\theta-}(x_t; t) + \frac{\sigma_t [f_{\theta-}(x_t; t) - f_{\theta-}(x_{\lambda t}; \lambda t)]}{\nu_t(t - \lambda t)} \right\|_2^2 \end{aligned}$$

which is identical to the objective of UCGM. When $\lambda = 0$, this reduces to the flow matching objective since $\lambda t = 0$ collapses $f_{\theta-}(x_0; 0) = x_0$. In this case, the objective becomes origin prediction, which in turn leads to F_θ with the velocity matching objective. Otherwise, setting $\lambda \rightarrow 1$ reduces the objective to consistency training by $\Delta = F_\theta(x_t; t) - F_{\theta-}(x_t; t) + \frac{\sigma_t}{\nu_t} \frac{df_{\theta-}(x_t; t)}{dt}$. For $\lambda \in (0, 1)$, the objective $\|f_\theta(x_t; t) - f_{\theta-}(x_{\lambda t}; \lambda t)\|_2^2$ yields consistency along the geometric sequence $\mathcal{T}_\lambda(t) = \{\lambda^k t\}_{k=0}^N$.

Define

$$g_t(x_t) = (x_t - \alpha_t f_t(x_t)) \sigma_t^{-1}$$

Then, when $f_t(x_t) = x$, it follows that $g_t(x_t) = z$ for $x_t = \alpha_t x + \sigma_t z$. Using this, the flow map can be formulated in a DDIM-like manner as

$$f_{t,s}(x_t) = \alpha_s f_t(x_t) + \sigma_s g_t(x_t) = \frac{\sigma_s}{\sigma_t} x_t + (\alpha_s - \frac{\sigma_s}{\sigma_t} \alpha_t) f_t(x_t)$$

Assuming the composition chain

$$f_{s,r}(f_{t,s}(x_t)) = f_{s,r}(\tilde{x}_s) = \frac{\sigma_r}{\sigma_s} \tilde{x}_s + (\alpha_r - \frac{\sigma_r}{\sigma_s} \alpha_s) f_s(\tilde{x}_s)$$

for $\tilde{x}_s = f_{t,s}(x_t)$, we obtain

$$f_{t,r}(x_t) - f_{s,r}(f_{t,s}(x_t)) = (\alpha_r - \frac{\sigma_r}{\sigma_s} \alpha_s) (f_t(x_t) - f_s(\tilde{x}_s))$$

For $s = \lambda^k t$ for some $k \in \mathbb{N}$, if it follows that $\tilde{x}_s = f_{t,s}(x_t) \approx x_s$, then $f_s(x_s) = f_t(x_t)$ and $f_{t,r} = f_{s,r} \circ f_{t,s}$. In this case, the flow map can be constructed along the geometric sequence $\mathcal{T}_\lambda(t)$.

In general, the velocity of the DDIM map is given by $\frac{d}{ds} f_{t,s} = \alpha'_s f_t + \sigma'_s g_t$. Since the unconditional velocity is

$$v_t^*(x_t) = \alpha'_t \mathbb{E}_{x|x_t}[x] + \sigma'_t \mathbb{E}_{x|x_t}[(x_t - \alpha_t x) \sigma_t^{-1}],$$

the DDIM map coincides with the flow map only when $v_s^*(f_{t,s}(x_t)) = \frac{d}{ds} f_{t,s}(x_t)$, which implies $\mathbb{E}_{x|\tilde{x}_s}[x] = f_t(x_t)$. Setting $s \rightarrow t$ reduces this condition to $\mathbb{E}_{x|x_t}[x] = f_t(x_t)$ by the identity assumption. However, this condition fails to preserve the injectivity of the flow map at $t = 1$ due to the mean collapse problem, resulting in a contradiction. Therefore, the DDIM-style map does not coincide with the flow map in general. $\square$

Reflow. Rectified flows introduce Reflow to straighten trajectories after training. In Reflow, sampling from the trained model is performed via

$$x_0 = x_1 + \int_1^0 v_\theta(x_t; t) dt \approx \text{ODESolver}(v_\theta, x_1, 1, 0)$$

followed by the finetune w.r.t. the coupling $\Pi_{Z,\theta} = p_Z(z)p_{v_\theta}(x|z)$. The velocity $\hat{v}_t$ of the trajectory induced by the coupling $\Pi_{Z,\theta}$ is given by

$$\hat{v}_t = x_1 - \left(x_1 + \int_1^0 v_\theta(x_t; t) dt \right) = \int_0^1 v_\theta(x_t; t) dt$$

which corresponds to the displacement of the flow map. Therefore, Reflow can be interpreted as direct supervision of the flow map under linear interpolation.

A.6. Suboptimality of direct training

Unlike Eulerian distillation, direct training does not guarantee convergence to the optimal flow map. Consider the direct training objective using the conditional velocity:

$$\mathcal{L}_{\text{DT}} = \mathbb{E}_{x,z,t,s} \left[\|\partial_t f_\theta(x_t; t, s) + v(x_t|x) \cdot \nabla_x f_\theta(x_t; t, s)\|_2^2 \right]$$

By defining the velocity error as $\Delta v = v_t(x_t|x) - v_t^*(x_t)$, we can rewrite the objective in the form of Eulerian Distillation:

$$\begin{aligned} \mathcal{L}_{\text{DT}} &= \mathbb{E}_{x,z,t,s} \left[\|\partial_t f_\theta(x_t; t, s) + (\Delta v + v_t^*(x_t)) \cdot \nabla_x f_\theta(x_t; t, s)\|_2^2 \right] \\ &= \mathbb{E}_{x,z,t,s} \left[\|\partial_t f_\theta(x_t; t, s) + \Delta v \cdot \nabla_x f_\theta(x_t; t, s) + v_t^*(x_t) \cdot \nabla_x f_\theta(x_t; t, s)\|_2^2 \right] \\ &= \mathbb{E}_{x,z,t,s} \left[\|\partial_t f_\theta(x_t; t, s) + v_t^*(x_t) \cdot \nabla_x f_\theta(x_t; t, s)\|_2^2 \right] + \mathbb{E}_{x,z,t,s} \left[\|\Delta v \cdot \nabla_x f_\theta(x_t; t, s)\|_2^2 \right] \end{aligned}$$

by the law of total variance since other terms are independent of x , and $\mathbb{E}_{x|x_t}[\Delta v] = 0$. In this case, the second term can be represented as

$$\mathbb{E}_{x,z,t,s} \left[\text{Var}_{x|x_t} [\Delta v \cdot \nabla_x f_\theta(x_t; t, s)] \right]$$

Under an independent coupling, the velocity error $\Delta v = v_t(x_t|x) - v_t^*(x_t)$ is typically nonzero. Consequently, unless $\|\nabla_x f_\theta(x_t; t, s)\|$ collapses to zero, the objective function inherently contains a larger variance term compared to that of Eulerian distillation. To minimize the overall loss, the optimizer faces a trade-off involving this variance. This introduces a bias that distorts the learned flow map towards becoming flatter by an external force $\Delta v_t = v_t - v_t^* \perp \nabla_x f_\theta(x_t; t, s)$.

Note that for flow matching, even when the loss term $\mathbb{E}_{x,z,t}[\|v(x_t|x) - F_\theta(x_t; t)\|_2^2]$ is decomposed as follows

$$\mathbb{E} \left[\|\Delta v + v^*(x_t) - F_\theta(x_t; t)\|_2^2 \right] = \mathbb{E} \left[\|v^*(x_t) - F_\theta(x_t; t)\|_2^2 \right] + \text{Var}[\Delta v]$$

The variance term, $\text{Var}[\Delta v]$, is independent of the network. Therefore, it does not affect convergence to the global optimum.

Euler-Lagrange Equation. Since the gradients of Eulerian distillation and direct training differ, the optima obtained by gradient-based optimization can also diverge. To find and compare the optima of the direct training objective, we apply the Euler-Lagrange equation. The objective can be represented in the vector form as

$$\mathcal{L}_{\text{DT}} = \mathbb{E}_{x,z,t,s} \left[\|\partial_t f_{t,s}(x_t) + v(x_t|x)^T \nabla_x f_{t,s}(x_t)\|_2^2 \right] = \iint_{\Omega} \rho_t(x_t) \mathbb{E}_{x,z,s|x_t} [\|\cdot\|_2^2] dx_t dt$$

We set the conditional expectation as the Lagrangian,

$$L(f, \partial_t f, \nabla_x f) = \mathbb{E}_{x,z,s|x_t} [\|\partial_t f_{t,s}(x_t) + v(x_t|x)^T \nabla_x f_{t,s}(x_t)\|_2^2]$$

The corresponding Euler-Lagrange equation is

$$\frac{\partial L}{\partial f} - \partial_t \left(\frac{\partial L}{\partial (\partial_t f)} \right) - \nabla_x \left(\frac{\partial L}{\partial (\nabla_x f)} \right) = 0 \iff \mathbb{E}_{x,z,s|x_t} [\partial_t R + \nabla \cdot (vR)] = 0$$

where the residue is defined as

$$R = \partial_t f_{t,s}(x_t) + v(x_t|x)^T \nabla_x f_{t,s}(x_t)$$

Letting $\Delta v = v(x_t|x) - v_t^*(x_t)$ and introducing the operator $L_* f = \partial_t f + (v_t^*)^T \nabla_x f$, the residue can be rewritten as $R = L_* f_{t,s} + \Delta v^T \nabla_x f_{t,s}$, so that

$$\begin{aligned} \mathbb{E}_{x|x_t} [\partial_t R] &= \partial_t \mathbb{E}_{x|x_t} [R] = \partial_t (L_* f_{t,s}) \\ \mathbb{E}_{x|x_t} [\nabla \cdot (vR)] &= \nabla \cdot \mathbb{E}_{x|x_t} [(v^* + \Delta v)(L_* f_{t,s} + \Delta v^T \nabla_x f_{t,s})] \\ &= \nabla \cdot (v^* L_* f_{t,s} + \mathbb{E}_{x|x_t} [\Delta v (\Delta v^T \nabla_x f_{t,s})]) \\ &= \nabla \cdot (v^* L_* f_{t,s} + \Sigma_{\Delta v|x_t} \nabla_x f_{t,s}) \end{aligned}$$

Therefore, the optimality condition becomes

$$\begin{aligned} \mathcal{EL} &= \mathbb{E}_{x,z,s|x_t} [\partial_t R + \nabla \cdot (vR)] \\ &= \partial_t (L_* f_{t,s}) + \nabla \cdot (v_t^* L_* f_{t,s}) + \nabla \cdot (\Sigma_{\Delta v|x_t} \nabla_x f_{t,s}) = 0 \end{aligned}$$

If we assume the L^2 -adjoint of L_* to be $L^* f = -\partial_t f - \nabla \cdot ((v_t^*)^T f)$, the condition simplifies to

$$L^* L_* f_{t,s} - \nabla \cdot (\Sigma_{\Delta v|x_t} \nabla_x f_{t,s}) = 0 \iff \|L_* f\|_2^2 + \nabla \cdot (\Sigma_{\Delta v|x_t} \nabla_x f_{t,s}) = 0$$

Thus, the optimum of the direct training arises precisely when the above condition is satisfied. When $\Sigma_{\Delta v|x_t} = \text{Cov}_{x|x_t}[v(x_t|x)] \rightarrow 0$, the condition reduces to $L_* f_{t,s} = 0$, which is equivalent to Eulerian distillation. In this case, the quadratic structure guarantees convergence through PSD curvature at the global optimum. Otherwise, we consider two cases separately: (i) $L_* f_{t,s} = 0$ and $\nabla \cdot (\Sigma_{\Delta v|x_t} \nabla_x f_{t,s}) = 0$, (ii) $L_* f_{t,s} \neq 0$ and $\nabla \cdot (\Sigma_{\Delta v|x_t} \nabla_x f_{t,s}) < 0$. In the second case, since the flow map is the unique solution of the Eulerian equation, $f_{t,s}$ no longer represents a valid flow map because it does not satisfy the equation. If this case is admitted, it corresponds to a suboptimal solution. Otherwise, if this case is excluded, the first case satisfying both conditions can hardly be regarded as a tractable or accessible solution. Therefore, the stable point of direct training either contains suboptimal solutions or remains practically inaccessible. $\square$

A.7. Instability of Consistency Training

The continuous-time consistency training objective employs a stop-gradient operation, ensuring that the main objective remains unchanged while making computation more efficient. In this case, the objective is defined as

$$\mathcal{L}_{\text{CT}} = \mathbb{E}_{x,z,t,s} \left[f_{t,s}(x_t)^T \frac{df_{t,s}^-(x_t)}{dt} \right]$$

where $f_{t,s}(x_t) = f_\theta(x_t; t, s)$ and detaching gradient is denoted by $f_{t,s}^-(x_t) = f_{\theta^-}(x_t; t, s)$.

$\mathcal{L}_{\text{CT}}$ reduces in value to Eulerian distillation even along the conditional trajectory, as demonstrated in Appendix A.5, paragraph on sCT. However, since the objective is expressed as a linear term, the Euler-Lagrange equation cannot determine a stationary point, as it contains no explicit terms of $f_{t,s}$ unless the gradient is detached:

$$\mathcal{EL} = \mathbb{E}_{x,z,s|x_t} [L f_{t,s}^- + v_t^T \nabla_x f_{t,s}^-] = 0$$

Moreover, while the quadratic term in Eulerian distillation ensures PSD curvature and provides stable convergence at the optimum, the consistency training objective does not guarantee convergence, as the Hessian vanishes and the curvature required to stabilize the optimum is absent. It only specifies the fixed point on $L f_{t,s} = 0$, and the gradient dynamics alone may fail to converge. $\square$

A.8. Suboptimality of Network-Induced Coupling

For an arbitrary coupling $(\hat{x}, \hat{z}) \sim \Pi_{X,Z}$ with $\hat{x}_t = \alpha_t \hat{x} + \sigma_t \hat{z}$, the gap between the conditional and marginal velocities is given by

$$\Delta v = \hat{x}'_t - v_t^*(\hat{x}_t) = \alpha'_t(\hat{x} - \mu_{x|\hat{x}_t}) + \sigma'_t(\hat{z} - \mu_{z|\hat{x}_t})$$

where $\mu_{x|x_t} = \mathbb{E}_{x|x_t}[x]$ and $\mu_{z|x_t} = \mathbb{E}_{z|x_t}[z]$ are conditional means. This follows since $v_t^*(x_t) = \mathbb{E}_{x|x_t}[v(x_t|x)]$ can be expressed as $\mathbb{E}_{x,z|x_t}[\alpha'_t x + \sigma'_t z]$. The general form of the loss can be expressed by

$$\mathbb{E}[\|A + g^T \Delta v\|_2^2] = \mathbb{E}[\|A\|_2^2] + 2\mathbb{E}[A^T g^T \Delta v] + \mathbb{E}[\|g^T \Delta v\|_2^2]$$

where $A = \partial_t f_{t,s}^\theta + g^T v_t^*(x_t)$ and $g = \nabla_x f_{t,s}^\theta$.

Case 1: Independent Coupling. In this setup, we use an independent coupling $(x, z) \sim p_{\text{data}} \times p_Z$ and the conditional velocity $v_t(x_t|x)$, which corresponds to setting $\hat{x} = x$ and $\hat{z} = z$. The velocity gap is $\Delta v = v_t(x_t|x) - v_t^*(x_t)$. The cross-term vanishes because the expectation of the gap is zero conditioned on x_t :

$$\mathbb{E}_{x,z|x_t}[\Delta v] = \mathbb{E}_{x,z|x_t}[v_t(x_t|x) - v_t^*(x_t)] = v_t^*(x_t) - v_t^*(x_t) = 0$$

Thus, $2\mathbb{E}[A \Delta v \cdot g] = 0$ and the loss simplifies to

$$\mathcal{L}_{\text{IC}} = \mathcal{L}_{\text{ED}} + \mathbb{E}[\|g^T \Delta v\|_2^2] = \mathcal{L}_{\text{ED}} + \mathbb{E}[\text{Var}_{x|x_t}[g^T \Delta v]]$$

The variance term, which represents the error from Eulerian distillation, can be expanded as:

$$\begin{aligned} \mathcal{E}_{\text{IC}} &= \text{Var}_{x,z|x_t}[g^T(\alpha'_t(x - \mu_{x|x_t}) + \sigma'_t(z - \mu_{z|x_t}))] \\ &= (\alpha'_t)^2 g^T \Sigma_{x|x_t} g + (\sigma'_t)^2 g^T \Sigma_{z|x_t} g + 2\alpha'_t \sigma'_t g^T \Sigma_{xz|x_t} g \end{aligned}$$

where $\Sigma_{x|x_t} = \text{Cov}_{x|x_t}(x, x)$, $\Sigma_{z|x_t} = \text{Cov}_{z|x_t}(z, z)$, and $\Sigma_{xz|x_t} = \text{Cov}_{x,z|x_t}(x, z)$.

Case 2: Generator-Induced Coupling. Generator-induced coupling methods replace one of the variables with a network prediction, $\hat{x} = f_{t,0}^\theta(x_t)$ with stop-gradient: $\hat{x}_t = \alpha_t f_{t,0}^\theta(x_t) + \sigma_t z$. For the first case, the velocity gap is:

$$\Delta v = \alpha'_t(f_{t,0}^\theta(x_t) - \mu_{x|\hat{x}_t}) + \sigma'_t(z - \mu_{z|\hat{x}_t})$$

The conditional expectation of the gap is no longer zero in general:

$$\mathbb{E}_{z|x_t}[\Delta v] = \alpha'_t(f_{t,0}^\theta(x_t) - \mu_{x|\hat{x}_t})$$

This introduces a non-zero cross-term in the loss, then the total error term for GC is:

$$\mathcal{E}_{\text{GC}} = 2\alpha'_t A^T g^T (f_{t,0}^\theta(x_t) - \mu_{x|\hat{x}_t}) + (\alpha'_t)^2 [g^T (f_{t,0}^\theta(x_t) - \mu_{x|\hat{x}_t})]^2 + (\sigma'_t)^2 g^T \Sigma_{z|x_t} g$$

Comparing the error terms, we find that $\mathcal{E}_{\text{IC}} > \mathcal{E}_{\text{GC}}$ if:

$$(\alpha'_t)^2 g^T \Sigma_{x|x_t} g + 2\alpha'_t \sigma'_t g^T \Sigma_{xz|x_t} g > 2\alpha'_t A^T g^T (f_{t,0}^\theta(x_t) - \mu_{x|\hat{x}_t}) + (\alpha'_t)^2 [g^T (f_{t,0}^\theta(x_t) - \mu_{x|\hat{x}_t})]^2$$

When the generator is a good estimator of the posterior mean, i.e., $f_{t,0}^\theta(x_t) \approx \mu_{x|\hat{x}_t}$, the right side becomes small. However, at $t = 1$, this condition reduced to $f_{1,0}^\theta(z) = \mu_{x|z} = \mu_X$, as shown in Appendix A.1, which results in posterior mean collapse. In this case, $f_{1,0}^\theta$ becomes constant as $\mathcal{E}_{\text{GC}}$ approaches zero, violating the injectivity required for a well-defined flow map, leading to a contradiction. Thus, generator-induced coupling reduces but cannot eliminate the error, preventing guaranteed convergence. It is also a suboptimal choice when continuous-time consistency training is employed.

VCT [31] introduces an additional network $g : x \mapsto z$ for $\hat{x} = x$ and $\hat{z} = g(x)$. In this case, enforcing $g(x) \approx \mu_{z|x}$ can reduce the gap, but does not eliminate the whole, since the term $(\alpha'_t)^2 g^T \Sigma_{x|x_t} g$ remains. From a consistency training perspective, since the independent coupling already exhibits a marginal velocity field, this becomes a negligible choice with respect to guaranteeing marginal velocity; however, it can reduce the variance of the loss when g is a good posterior approximator. $\square$

A.9. Linearization Cost

Recall that $f_{t,s}^\theta(x_t) = \nu^{-1}(A'_{t,s}x_t - A_{t,s}F_{t,s}^\theta(x_t))$ for $A_{t,s} = \alpha_s\sigma_t - \sigma_s\alpha_t$. Under the given formulation of flow maps, a pseudo velocity network F_θ learns the weighted displacement between x_t and x_s . From the perspective of F_θ , this process can be seen as shifting the training target from a classical instantaneous velocity field to an average velocity field (a weighted displacement field), which transforms the path between x_t and x_s into the linear form of x_t and F_θ . We term it *linearization*.

Differentiating with respect to the starting point t gives

$$\frac{df_\theta(x_t; t, s)}{dt} = \nu^{-1} \left(A''_{t,s}x_t + A'_{t,s}(v_t^*(x_t) - F_\theta(x_t; t, s)) - A_{t,s} \frac{dF_\theta(x_t; t, s)}{dt} \right)$$

Following sCT [23], the gradient of the consistency training can be written as

$$\begin{aligned} & \nabla_\theta \mathbb{E} \left[2f_\theta^T(x_t; t, s) \frac{df_{\theta-}(x_t; t, s)}{dt} \right] \\ & \propto \nabla_\theta \mathbb{E} \left[-A_{t,s} \nu^{-2} F_\theta(x_t; t, s) \cdot \left(A''_{t,s}x_t + A'_{t,s}(v_t^*(x_t) - F_{\theta-}(x_t; t, s)) - A_{t,s} \frac{dF_{\theta-}(x_t; t, s)}{dt} \right) \right] \end{aligned}$$

In this case, $v_t^* - F_\theta$ can be interpreted as the flow matching term weighted by $A'_{t,s}$, and $A'_{t,s}x_t - A_{t,s}dF_\theta/dt$ as a linearization term involving the JVP, which penalizes the t -dependent outputs of f_θ . For a linear interpolation, $A_{t,s}$ takes the form $A_{t,s} = t - s$, while for a trigonometric interpolation $A_{t,s} = \sin(t - s)$, both are proportional to $t - s$. As $s \rightarrow t$ and $(t - s) \rightarrow 0$, the contribution of the linearization term vanishes, while the flow matching term is amplified. Conversely, as $s \rightarrow 0$, the linearization term is amplified and the flow matching term diminishes.

We note that the linearization cost increases with step size, making optimization more challenging. This is because the linearization term involves a complex structure induced by the JVP, whereas the flow matching term requires only a simple forward pass. Such a complex structure may introduce additional fixed points, hindering convergence to the desired optimum.

A.10. Training-time Classifier-free Guidance

MeanFlow suggests Classifier-free Guidance [13] (CFG) for flow maps:

$$\begin{aligned} & \nabla_\theta \mathbb{E} [\|F_\theta(x_t; t, s, c) - \text{sg}[\tilde{v}_t(x_t|x) - (t - s)(\tilde{v}_t(x_t|x) \cdot \nabla_x F_\theta + \partial_t F_\theta)]\|] \\ & \text{where } \tilde{v}_t(x_t|x) = F_{\theta-}(x_t; t, \emptyset) + \omega(v_t(x_t|x) - F_{\theta-}(x_t; t, \emptyset)) \end{aligned}$$

for a conditional class label c , the null class label $\emptyset$ of unconditional generation, and a guidance scale ω . We find that this formulation guarantees that the flow map follows the ground-truth CFG trajectory. The ground-truth CFG trajectory is given by $\tilde{v}_t^*(x_t; c) = v_t^*(x_t) + \omega(v_t^*(x_t; c) - v_t^*(x_t))$ where $v_t^*(x_t; c)$ is the class-conditional marginal velocity $v_t^*(x_t; c) = \mathbb{E}_{x|x_t, c}[v_t(x_t|x)]$ and $v_t^*(x_t)$ is the marginal velocity $v_t^*(x_t) = \mathbb{E}_{x|x_t}[v_t(x_t|x)] = \mathbb{E}_c[v_t^*(x_t; c)]$.

To prove this, we first consider only the flow matching setting:

$$\begin{aligned} & \nabla_\theta \mathbb{E} [\|F_\theta(x_t; t, c) - \tilde{v}_t(x_t|x)\|_2^2] \\ & = \nabla_\theta \mathbb{E} [\|F_\theta(x_t; t, c) - (F_{\theta-}(x_t; t, \emptyset) + \omega(v_t(x_t|x) - F_{\theta-}(x_t; t, \emptyset)))\|_2^2] \end{aligned}$$

With label dropout $c = \emptyset$, the objective reduces to weighted unconditional flow matching that converges to the unconditional marginal velocity:

$$\begin{aligned} & \nabla_\theta \mathbb{E} [\|F_\theta(x_t; t, \emptyset) - (F_{\theta-}(x_t; t, \emptyset) + \omega(v_t(x_t|x) - F_{\theta-}(x_t; t, \emptyset)))\|_2^2] \\ & = \mathbb{E} [2\omega \cdot \nabla_\theta F_\theta(x_t; t, \emptyset) \cdot (F_{\theta-}(x_t; t, \emptyset) - v_t(x_t|x))] \\ & = \nabla_\theta \mathbb{E} [\omega \|F_\theta(x_t; t, \emptyset) - v_t(x_t|x)\|_2^2] \end{aligned}$$

In this case, we can set $F_\theta(x_t; t, \emptyset) \approx v_t^*(x_t)$ if the network has sufficient capacity. For the conditional case $c \neq \emptyset$, applying

the tower property yields:

$$\begin{aligned}
& \nabla_{\theta} \mathbb{E} \left[\|F_{\theta}(x_t; t, c) - (F_{\varnothing}(x_t; t) + \omega(v_t(x_t|x) - F_{\varnothing}(x_t; t)))\|_2^2 \right] \\
&= \nabla_{\theta} \mathbb{E} \left[\mathbb{E}_{x|x_t, c} \left[\|F_{\theta}(x_t; t, c) - (F_{\varnothing}(x_t; t) + \omega(v_t(x_t|x) - F_{\varnothing}(x_t; t)))\|_2^2 \right] \right] \\
&= \nabla_{\theta} \mathbb{E} \left[\mathbb{E}_{x|x_t, c} \left[\|F_{\theta}(x_t; t, c) - (F_{\varnothing}(x_t; t) + \omega(v_t^*(x_t; c) - v_t^*(x_t; c) + v_t(x_t|x) - F_{\varnothing}(x_t; t)))\|_2^2 \right] \right] \\
&= \nabla_{\theta} \mathbb{E} \left[\mathbb{E}_{x|x_t, c} \left[\|F_{\theta}(x_t; t, c) - (F_{\varnothing}(x_t; t) + \omega(v_t^*(x_t; c) - F_{\varnothing}(x_t; t))) + \omega(v_t^*(x_t; c) - v_t(x_t|x))\|_2^2 \right] \right] \\
&\approx \nabla_{\theta} \mathbb{E} \left[\|F_{\theta}(x_t; t, c) - (v_t^*(x_t) + \omega(v_t^*(x_t; c) - v_t^*(x_t)))\|_2^2 \right] + \omega^2 \cdot \nabla_{\theta} \mathbb{E} [\text{Var}_{x|x_t, c} [v_t(x_t|x)]] \\
&= \nabla_{\theta} \mathbb{E} \left[\|F_{\theta}(x_t; t, c) - \tilde{v}_t^*(x_t; c)\|_2^2 \right]
\end{aligned}$$

where $F_{\varnothing}(x_t; t) = F_{\theta^-}(x_t; t, \varnothing)$ for brevity. Thus, the self-referential target $\tilde{v}_t(x_t|x)$ with label dropout ensures convergence to the ground-truth CFG velocity.

Consider the convergence of the flow map to the CFG velocity field. Following Appendix A.5, let $L_* f_{t,s} = \partial_t f_{t,s} + \tilde{v}_t^* \cdot \nabla_x f_{t,s}$ and $\Delta v = \tilde{v}_t(x_t|x) - \tilde{v}_t^*(x_t; c)$. It is sufficient to show that $\mathbb{E}_{x|x_t, c} [\Delta v] = 0$, and it is guaranteed when label dropout is applied and the flow matching objective is jointly optimized. Thus, the objective guarantees convergence to the flow map induced by the CFG velocity. $\square$

B. improved Self-Distillation

B.1. Guarantee the Convergence

Revisit our objective

$$\mathcal{L} = \mathbb{E}_{x,z,t,s} \left[\|F_{t,t}^{\theta}(x_t; t, t) - v_t(x_t|x)\|_2^2 + \left\| \partial_t f_{t,s}^{\theta}(x_t) + F_{t,t}^{\theta^-}(x_t) \cdot \nabla_x f_{t,s}^{\theta}(x_t) \right\|_2^2 \right]$$

The first term of the right side trains $F_{t,t}^{\theta}(x_t)$ to approximate the marginal velocity via the flow matching objective, while the second term learns the flow map $f_{t,s}$ along the trajectory of $F_{t,t}^{\theta^-}$ in a self-distillation manner.

Individually, each term is guaranteed to converge to its desired optimum, the marginal velocity and the flow map of the velocity $F_{t,t}^{\theta^-}$ by Proposition 1 (Appendix A.3). From a joint perspective, we need to consider $t = s$, since the network is forced to optimize both terms simultaneously at this point. As Eulerian distillation collapses to the flow matching objective when $t \rightarrow s$, the second term trains the model to learn the instantaneous velocity of the trajectory (Appendix A.9). In this case, $F_{t,t}^{\theta}$ learns from $F_{t,t}^{\theta^-}$, and inductively approximates $v_t^*(x_t)$ through the first term. This naturally reduces to a non-conflict joint training. For $t \neq s$, the network is conditioned differently in the two terms, and it can learn the proper mapping provided that the network capacity is sufficient. Consequently, the overall objective trains the network to follow the marginal velocity as the trajectory of the flow map naturally. $\square$

B.2. Deriving Final Objective

Recall the consistency training objective under the generalized flow map (Appendix A.9):

$$\begin{aligned}
& \nabla_{\theta} \mathbb{E} \left[2f_{\theta}^T(x_t; t, s) \frac{df_{\theta^-}(x_t; t, s)}{dt} \right] \\
& \propto \nabla_{\theta} \mathbb{E} \left[-A_{t,s} \nu^{-2} F_{\theta}(x_t; t, s) \cdot \left(A_{t,s}' x_t + A_{t,s}' (v_t^*(x_t) - F_{\theta^-}(x_t; t, s)) - A_{t,s} \frac{dF_{\theta^-}(x_t; t, s)}{dt} \right) \right] \\
& = A_{t,s} \nu^{-2} \cdot \nabla_{\theta} \mathbb{E} [\|F_{\theta}(x_t; t, s) - \text{sg}[F_{\text{tgt}}(x_t; t, s)]\|_2^2]
\end{aligned}$$

where $F_{\text{tgt}}(x_t; t, s) = F_{\theta}(x_t; t, s) + \left(A_{t,s}' x_t + A_{t,s}' (v_t^*(x_t; t) - F_{\theta}(x_t; t, s)) - A_{t,s} \frac{dF_{\theta}(x_t; t, s)}{dt} \right)$.

To follow the marginal velocity, we replace $v_t^*(x_t)$ with instantaneous velocity $F_{t,t}^{\theta}(x_t)$ while jointly training with $\mathcal{L}_{\text{CFM}}$. Particularly, for linear interpolation, we have $A_{t,s} = t-s$, $A_{t,s}' = 1$ and $A_{t,s}'' = 0$. This simplifies the target to $F_{\text{tgt}}^{\text{lin}}(x_t; t, s) = v_t^*(x_t) - (t-s)F_{\theta}'(x_t; t, s)$ which coincides with the regression target of MeanFlow. For trigonometric interpolation, we

have $A_{t,s} = \sin(t-s)$, $A'_{t,s} = \cos(t-s)$ and $A''_{t,s} = -\sin(t-s)$. Thus, the target becomes $F_{\text{tgt}}^{\text{tri}} = F_{\theta}(x_t; t, s) + \cos(t-s) \cdot (v_t^*(x_t) - F_{\theta}(x_t; t, s)) - \sin(t-s) \cdot (x_t + F'_{\theta}(x_t; t, s))$. 1135

Although consistency training already guarantees the marginal flow map at its fixed point, the gradient in practice can exhibit a gap expressed as: 1136

$$\mathbb{E}_{x,z,t,s}[f_{t,s}^{\theta}(\Delta v \cdot \nabla_x f_{t,s}^{\theta})] 1137$$

When self-distillation is combined with flow matching, $\mathbb{E}[\|F_{\theta}(x_t; t, t) - v(x_t|x)\|_2^2]$, the velocity error $\Delta v = F_{\theta}(x_t; t, t) - v_t^*(x_t)$ can be further reduced compared to $\Delta v = v_t(x_t|x) - v_t^*(x_t)$, thereby stabilizing the training. 1138

For JVP approximation, to ensure that dx_t/dt follows the velocity $v_{\theta}(x_t; t) = F'_{t,t}(x_t)$, we approximate 1139

$$\frac{dx_t}{dt} \approx \frac{[x_t + \epsilon \cdot v_{\theta}(x_t; t)] - [x_t - \epsilon \cdot v_{\theta}(x_t; t)]}{2\epsilon} = v_{\theta}(x_t; t) 1140$$

Thus, the full JVP approximation becomes 1141

$$\frac{dF_{\theta}(x_t; t, s)}{dt} = \frac{F_{\theta}(x_t + \epsilon \cdot v_{\theta}(x_t; t), t + \epsilon, s) - F_{\theta}(x_t - \epsilon \cdot v_{\theta}(x_t; t), t - \epsilon, s)}{2\epsilon} + O(\epsilon^2) 1142$$

Applying adaptive weighting, our final objective is 1143

$$\begin{aligned} \tilde{\mathcal{L}}_{t,s}(x_t, x) &= \|F_{\theta}(x_t; t, t) - v_t(x_t|x)\|_2^2 + \|F_{\theta}(x_t; t, s) - \text{sg}[F_{\text{tgt}}(x_t; t, s)]\|_2^2 \\ \mathcal{L}_{\text{iSD}} &= \mathbb{E}_{x,z,t,s} [w_{t,s}(x_t, x) \cdot \tilde{\mathcal{L}}_{t,s}(x_t, x)]; \text{ where } w_{t,s}(x_t, x) = (\text{sg}[\tilde{\mathcal{L}}_{t,s}(x_t, x)] + \eta)^{-p} \end{aligned} 1144$$

When using JVP approximation, the resulting asymptotic error is bounded by $O(\epsilon^2)$. This error appears acceptable under BF16 precision, as the approximation shows better results than direct JVP computation. The complete training and sampling algorithms are provided in Algorithm 1 and Algorithm 2. 1145

Algorithm 1 (iSD Training) Training algorithm of vanilla iSD

Require: Noise distribution p_Z , data distribution p_X , model F_{θ} , learning rate μ , time distribution τ , adaptive weighting (η, p) , JVP approximation step size ϵ , class labels c

Ensure:

- 1: **repeat**
 - 2: $z \sim p_Z, \quad x \sim p_X, \quad t, s \leftarrow \tau$
 - 3: $x_t \leftarrow \alpha_t x + \sigma_t z, \quad v_t \leftarrow \alpha'_t x + \sigma'_t x$
 - 4: $F_{t,s} \leftarrow F_{\theta}(x_t; t, s, c), \quad F_{t,t} \leftarrow F_{\theta}(x_t; t, t, c)$
 - 5: $F'_{t,s} \leftarrow [F_{\theta}(x_t + \epsilon F_{t,t}; t + \epsilon, s, c) - F_{\theta}(x_t - \epsilon F_{t,t}; t - \epsilon, s, c)] / (2\epsilon)$ ▷ JVP-Approx.
 - 6: $F_{\text{tgt}} \leftarrow F_{t,s} + (A''_{t,s} x_t + A'_{t,s}(F_{t,t} - F_{t,s}) - A_{t,s} F'_{t,s})$
 - 7: $\tilde{\mathcal{L}}_{t,s} \leftarrow \|F_{t,t} - v_t\|_2^2 + \|F_{t,s} - \text{sg}[F_{\text{tgt}}]\|_2^2$ ▷ Optimization Target
 - 8: $\mathcal{L} \leftarrow \tilde{\mathcal{L}}_{t,s} \times (\text{sg}[\tilde{\mathcal{L}}_{t,s}] + \eta)^{-p}$ ▷ Adaptive Weighting
 - 9: $\theta \leftarrow \theta - \mu \nabla_{\theta} \mathcal{L}$ ▷ Model Update
 - 10: **until** Convergence
-

Algorithm 2 (iSD Sampling) Sampling algorithm of vanilla iSD

Require: Initial noise $z \sim p_Z$, trained model F_{θ} , class labels c , sampling time steps $\{t_i\}_{i=1}^N$

Ensure:

- 1: $x \leftarrow z$
 - 2: **for** $i \leftarrow 1$ to N **do**
 - 3: $x \leftarrow \cos(t_{i+1} - t_i) \cdot x + \sin(t_{i+1} - t_i) \cdot F_{\theta}(x; t_i, t_{i+1}, c)$
 - 4: **end for**
 - 5: **return** x
-

B.3. Random initialization instead of Preconditioners

When constructing the iSD framework, we focus on random initializations rather than preconditioners. As shown in Tab. 2, pretrained networks with better FIDs do not necessarily yield better FIDs in the resulting consistency models. With random initialization, one can explore different random seeds and retrain the network to obtain improved results. There also remains potential to improve performance using alternative initialization strategies, such as leveraging preconditioners. However, when adopting preconditioners as baselines, it becomes difficult to systematically design experiments to discover *better preconditioners*: which data, architectures, and hyperparameters should be used, how to quantify the quality of preconditioners, and whether perturbing pretrained weights is necessary. Therefore, we conclude that random initialization makes the reproducibility problem more tractable, and we construct the iSD framework as a reproducible, randomly initializable consistency training scheme.

B.4. Classifier-free Guidance of Flow Map Models

Pre-CFG. By abstracting the guiding trajectory to v_θ , the flow map model can naturally be trained to follow the specific trajectory as long as it is Lipschitz continuous. To achieve this, we replace the guiding trajectory and refer to this training scheme as *Pre-CFG*. We consider three variants of objectives: iSD-C, iSD-U, and iSD-T.

iSD-U (Guidance-Unconditional). Given a label c and a null class label $\emptyset$, let the corresponding velocity fields be $F_{t,t}(x_t; c)$ and $F_{t,t}(x_t; \emptyset)$. If both are globally Lipschitz continuous, then the CFG trajectory $\tilde{v}_\theta(x_t; t, c) = \tilde{v}_t(x_t; c)$ is also globally Lipschitz continuous, since any linear combination of Lipschitz continuous functions remains Lipschitz continuous.

$$\tilde{v}_\theta(x_t; t, c) = F_\theta(x_t; t, t, \emptyset) + \omega(F_\theta(x_t; t, t, c) - F_\theta(x_t; t, t, \emptyset))$$

Thus, the flow map can be trained to follow the CFG velocity field. As a naive approach, we first train the network to align with $\tilde{v}_\theta$, while leaving the remaining components unchanged.

iSD-C (Guidance-Conditional). In this case, however, we need to address a conflict: $\mathcal{L}_{\text{CFM}}$ enforces $F_{t,t} \approx v_t^*$ while $\mathcal{L}_{\text{SD-R}}$ enforces $F_{t,t} \approx \tilde{v}_\theta$. To resolve this, we append the guidance scale ω as an additional condition, $F_{t,t}^\theta(x_t; c, \omega)$. Then, the modified objectives are given by:

$$\begin{aligned}\mathcal{L}_{\text{CFM}} &= \mathbb{E} [\|F_\theta(x_t; t, t, c, 1.0) - v_t(x_t|x)\|_2^2] \\ \mathcal{L}_{\text{iSD-C}} &= \mathbb{E} [\|F_\theta(x_t; t, s, c, \omega) - \text{sg}[F_{\text{tgt}}(x_t; t, s, c, \omega)]\|_2^2]\end{aligned}$$

where

$$\begin{aligned}F_{\text{tgt}} &= F_{t,s}(x_t; c, \omega) + (A''_{t,s}x_t + A'_{t,s}(\tilde{v}_t(x_t; c) - F_{t,s}(x_t; c, \omega)) - A_{t,s}F'_{t,s}(x_t; c, \omega)) \\ \tilde{v}_t(x_t; c) &= F_{t,t}(x_t; \emptyset, 1.0) + \omega(F_{t,t}(x_t; c, 1.0) - F_{t,t}(x_t; \emptyset, 1.0))\end{aligned}$$

with θ omitted for brevity. Hence, $\mathcal{L}_{\text{CFM}}$ ensures $F_{t,t}(x_t; c, 1.0) \approx v_t^*(x_t; c)$, while $\mathcal{L}_{\text{iSD-C}}$ ensures $F_{t,t}(x_t; c, \omega) \approx \tilde{v}_t(x_t; c)$. This choice is natural, as $\tilde{v}_t = v_t^*$ when $\omega = 1$.

iSD-T (Training-time CFG). To resolve the conflict from another perspective, we adopt the training-time CFG described in Appendix A.10. As proven there, the training target $\tilde{v}_t(x_t|x) = (1 - \omega)F_{\theta-}(x_t; t, t, \emptyset) + \omega v_t(x_t|x)$ guarantees the convergence of flow matching networks to the CFG velocity field. Accordingly, we modify the flow matching objective to follow the CFG velocity, distilling the approximated velocity into flow mappings:

$$\begin{aligned}\mathcal{L}_{\text{CFM}} &= \mathbb{E} [\|F_\theta(x_t; t, t, c) - \tilde{v}_t(x_t|x)\|_2^2] \\ \mathcal{L}_{\text{iSD-T}} &= \mathbb{E} [\|F_\theta(x_t; t, s, c) - \text{sg}[F_{\text{tgt}}(x_t; t, s, c)]\|_2^2]\end{aligned}$$

where $v_\theta(x_t; t) = F_\theta(x_t; t, t, c)$. In this case, the network approximates the CFG velocity, and the resulting flow mappings naturally follow the CFG trajectory. Unlike $\mathcal{L}_{\text{iSD-U}}$, the network does not compromise theoretical guarantees at $s = t$. We finalize our method by adopting iSD-T as the default configuration. The ablation studies are provided in Appendix C.3, and the training algorithm is described in Algorithm 3.

Post-CFG. However, *Post-CFG* defined as

$$\tilde{F}_\theta(x_t; t, s, c) = F_\theta(x_t; t, s, \emptyset) + \omega(F_\theta(x_t; t, s, c) - F_\theta(x_t; t, s, \emptyset))$$

does not follow the CFG trajectory. This discrepancy arises from the definition of the flow map,

$$f_\theta(x_t; t, s) = x_t + \int_t^s v_\tau^*(x_\tau) d\tau$$

which performs the path integral along a specific trajectory induced by v_τ^* . For a CFG trajectory, the path integral should be taken along $\tilde{v}_\tau$. In contrast, Post-CFG computes two separate forward passes, integrating along $v_\tau^*(x_\tau; c)$ and $v_\tau^*(x_\tau; \emptyset)$, rather than along $\tilde{v}_\tau$. As a result, the integration differs from the expected CFG trajectory. The detailed procedure of Post-CFG follows Algorithm 4.

Algorithm 3 (iSD-T Training) Training algorithm of iSD-T

Require: Noise distribution p_Z , data distribution p_X , model F_θ , learning rate μ , time distribution τ , adaptive weighting (η, p) , JVP approximation ϵ , Pre-CFG scale ω , class labels c .

Ensure:

- 1: **repeat**
 - 2: $z \sim p_Z, \quad x \sim p_X, \quad t, s \leftarrow \tau$
 - 3: $x_t \leftarrow \alpha_t x + \sigma_t z, \quad v_t \leftarrow \alpha'_t x + \sigma'_t z$
 - 4: $\tilde{v}_t = \omega v_t + (1 - \omega) F_\theta(x_t; t, \emptyset)$
 - 5: $F_{t,s} \leftarrow F_\theta(x_t; t, s, c), \quad F_{t,t} \leftarrow F_\theta(x_t; t, t, c)$
 - 6: $F'_{t,s} \leftarrow [F_\theta(x_t + \epsilon F_{t,t}; t + \epsilon, s, c) - F_\theta(x_t - \epsilon F_{t,t}; t - \epsilon, s, c)] / (2\epsilon)$ ▷ JVP-Approx.
 - 7: $F_{\text{tgt}} \leftarrow F_{t,s} + (A''_{t,s} x_t + A'_{t,s} (F_{t,t} - F_{t,s}) - A_{t,s} F'_{t,s})$
 - 8: $\tilde{\mathcal{L}}_{t,s} \leftarrow \|F_{t,t} - \tilde{v}_t\|_2^2 + \|F_{t,s} - \text{sg}[F_{\text{tgt}}]\|_2^2$ ▷ Optimization Target
 - 9: $\mathcal{L} \leftarrow \tilde{\mathcal{L}}_{t,s} \times (\text{sg}[\tilde{\mathcal{L}}_{t,s}] + \eta)^{-p}$ ▷ Adaptive Weighting
 - 10: $\theta \leftarrow \theta - \mu \nabla_\theta \mathcal{L}$ ▷ Model Update
 - 11: **until** Convergence
-

Algorithm 4 (Post-CFG Sampling) Sampling algorithm of iSD with Post-CFG

Require: Initial noise $z \sim p_Z$, model F_θ , Post-CFG scale ω , class labels c , sampling steps $\{t_i\}_{i=1}^N$.

Ensure:

- 1: $x \leftarrow z$
 - 2: **for** $i \leftarrow 1$ to N **do**
 - 3: $\tilde{F}_{t,s} \leftarrow (1 - \omega) F_\theta(x; t_i, t_{i+1}, \emptyset) + \omega F_\theta(x; t_i, t_{i+1}, c)$ ▷ Post-CFG
 - 4: $x \leftarrow \nu^{-1}(A'_{t_i, t_{i+1}} x - A_{t_i, t_{i+1}} \tilde{F}_{t,s})$
 - 5: **end for**
 - 6: **return** x
-

C. Experimental Details

C.1. Reproducibility of Consistency Training

To evaluate the reproducibility of consistency training, we conducted experiments within the UCGM [38] framework. We compared the FID scores of several models trained under different initialization conditions. Following UCGM, we first extract latent representations of ImageNet-1K 256×256 using VAAE [41]. All few-step models were trained with the same hyperparameters and settings: RAdam optimizer with a learning rate of 0.0001, weight decay of 0.0, $\beta_1 = 0.9$, $\beta_2 = 0.999$, batch size of 1024, gradient clipping at 0.1, and timestep t sampled from Beta(0.8, 1.0). For enhancement, we applied a label drop ratio of 0.1, an enhancement range of (0, 0.75), and an enhancement ratio of 2.0. We also used the cosine function as the loss weighting function and trained all models with linear interpolation for 40K iterations.

Multistep Baseline. We trained the DiT-XL/1 architecture initialized from the publicly released multistep checkpoint of UCGM. This configuration achieved a 2-step FID of 2.52, which is reasonable but still falls short of the reported FID of 1.42.

LightningDiT. We trained the LightningDiT-XL/1 architecture from its released pretrained model. In this setting, the model achieved a 2-step FID of 9.59, which is worse than the reported FID.

In-house Multistep Model. In this experiment, we trained the DiT-XL/2 architecture from scratch. For training, we used AdamW [22] with a learning rate of 0.0002, $\beta_1 = 0.9$, $\beta_2 = 0.95$, EMA decay weight of 0.999, and timestep t sampled from Beta(1, 1). We used an enhancement ratio of 0.47 and a cosine weighting function. After training the multistep model for 800k iterations, we trained a few-step model initialized from it using consistency training under the same few-step settings. This resulted in a 2-step FID of 5.78.

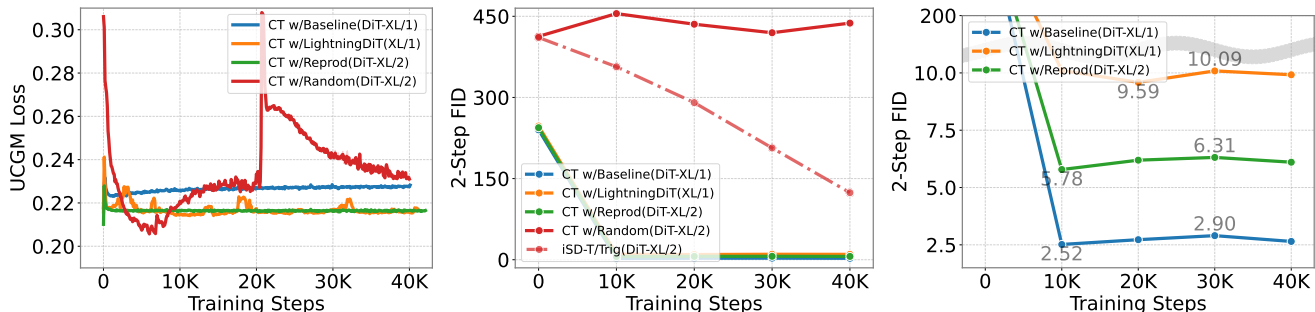


Figure 4. Training curves of consistency training. (a) Training loss over training steps. (b) Two-step FIDs over training steps. (c) Rescaled Y-axis for the consistency model with a preconditioner.

Without Preconditioner. We train a DiT-XL/2 architecture from randomly initialized weights without any preconditioner. In this case, training consistently failed, with the loss diverging and no meaningful samples being generated. While 40K steps may appear insufficient for scratch training, other scratch training methods already show a rapid decrease by 40K steps (Fig. 4). We consider that this is enough to check the unstable dynamics compared to other models.

These experiments suggest that consistency training is highly sensitive to initialization and the choice of preconditioner. It proves unstable under random initialization and requires a well-trained multistep baseline for stable optimization. Furthermore, even when initialized from a pretrained model, consistency training demonstrates limited robustness and reproducibility across different architectures and setups.

C.2. Implementation

ImageNet 256×256. SD-VAE [27] was used to encode images from the ImageNet 256 dataset into a $32 \times 32 \times 4$ latent representation. For DiT [24] models, we employed RMSNorm [44], QK normalization [11], and RoPE [37] for minor improvements. Each model was scaled by depth and hidden dimension, while the patch size was kept fixed. We sampled t and s independently from Beta(0.8, 1.0), and set $t, s := \max(t, s), \min(t, s)$. For trigonometric interpolation, we additionally scale the timesteps as $t \leftarrow \frac{\pi}{2}t$ and $s \leftarrow \frac{\pi}{2}s$. For generations, we uniformly sampled the intermediate timesteps from the interval without additional engineering.

In ablation studies, DiT-B/4 was trained for 400K steps with a batch size of 256 and BF16 precision, on three distinct GPUs to evaluate reproducibility: RTX 3090, RTX A6000, and A100. Since direct JVP computation requires approximately 25GB VRAM to train DiT-B/4 at this batch size, we apply two-step gradient accumulation on RTX 3090 GPUs. Since the required VRAM was reduced to 18GB with the JVP approximation, gradient accumulation was not used in this setting. For iSD-T with our final setting, the corresponding wall-clock times are 88 GPU hours on an RTX 3090, 67 GPU hours on an A6000, and 32 GPU hours on an A100.

In scalability experiments, DiT-B/2 and DiT-XL/2 were trained for 400K steps, and DiT-XL/2 was further trained up to 800K steps for comparison with prior work, where the corresponding wall-clock time is 216 A100 GPU hours. For further improvements, we tune some hyperparameters for DiT-XL/2, denoted by DiT-XL/2[†], and reported results in Tab. 5 are from DiT-XL/2[†]. Detailed training parameters are provided in Tab. 6.

Table 6. Experimental Settings

Dataset	ImageNet 256×256					CIFAR-10
Preprocessor	SD-VAE [27]					Identity
Input size	$32 \times 32 \times 4$					$32 \times 32 \times 3$
Backbone	DiT-B/4	DiT-B/2	DiT-L/2	DiT-XL/2	DiT-XL/2 [†]	UNet+ [35]
Params (M)	131	131	459	676	676	56
Depth	12	12	24	28	28	-
Hidden dim	768	768	1024	1152	1152	-
Heads	12	12	16	16	16	-
Patch size	4×4	2×2	2×2	2×2	2×2	1×1
Dropout	0.0					0.2
Interpolation	Trigonometric					Trigonometric
Self-distillation	$\mathcal{L}_{\text{iSD-T}}$					$\mathcal{L}_{\text{iSD}}$
Joint training	Enabled					Enabled
JVP	Approximation: Eq. (16)					<code>torch.func.jvp</code>
ϵ	0.005					-
p	1.0					0.75
η	0.01					0.01
Pre-CFG ω	3.0					-
FP Precision	BF16		BF16		FP32	
Batch size	256		256		256	
Label dropout	0.1		0.1		-	
Optimizer	AdamW [22]		AdamW		AdamW	
LR Scheduler	Constant		Constant		Linear Warmup	
β_1	0.9		0.9		0.9	
β_2	0.999		0.9995		0.999	
Learning rate	$1\text{e-}4$		$5\text{e-}5$		$1\text{e-}3$	
Training steps	400K		800K		950K	
Weight decay	0		0		0	
EMA decay	0.99995		0.99995		0.99995	

CIFAR-10. For CIFAR-10, we train the model in pixel space without any VAE latent encoders. The model is based on UNet+ [35] without class conditioning. It is trained for 950K steps on two RTX A6000 GPUs with a global batch size of 256, requiring approximately 15 days of wall-clock time (715 A6000 GPU hours). Non-leaky data augmentation [16] is also applied, excluding vertical flipping and rotation. Further details of the experimental settings can be found in Tab. 6.

C.3. Additional Experiments

JVP Approximation. To validate the effectiveness of JVP approximation, we not only apply the JVP approximation to iSD, but also to MeanFlow. With the direct JVP, the training requires approximately 25GB of VRAM and 58 A100 GPU hours under the ablation setting. After applying the approximation, the VRAM requirement reduces to 11GB and the wall-clock time to 21.5 GPU hours. The FID also decreases from 14.83 to 10.74, demonstrating a consistent decreasing trend.

Variants of Pre-CFG. In the ablation setting, as shown in Fig. 5a and Tab. 9, applying Pre-CFG $\mathcal{L}_{\text{iSD-U}}$ with $\omega = 1.5$ improves FIDs compared to vanilla $\mathcal{L}_{\text{iSD}}$. However, training diverges when $\omega > 3.0$. This instability can be attributed to the conflict discussed in Appendix B.4, which disrupts optimal convergence as the guidance scale increases.

When the guidance scale is appended as a condition, $\mathcal{L}_{\text{iSD-C}}$ enables training at higher guidance scales, outperforming $\mathcal{L}_{\text{iSD}}$. However, $\mathcal{L}_{\text{iSD-C}}$ consistently underperforms $\mathcal{L}_{\text{iSD-U}}$. Intuitively, the additional condition forces the network to learn both CFG and non-CFG mappings, imposing an extra burden on the network. Even though $\mathcal{L}_{\text{iSD-U}}$ compromises the theoretical guarantees at $s = t$, few-step generation commonly assumes $s \ll t$, resulting in comparable performance to iSD-T and making the issue negligible in practice. On the other hand, iSD-T neither introduces the conflict nor imposes additional burden on the network. It consistently outperforms both variants. Therefore, we adopt iSD-T as the default.

We also examined Post-CFG, as illustrated in Fig. 5b. iSD-U shows improved results when Post-CFG is applied, par-

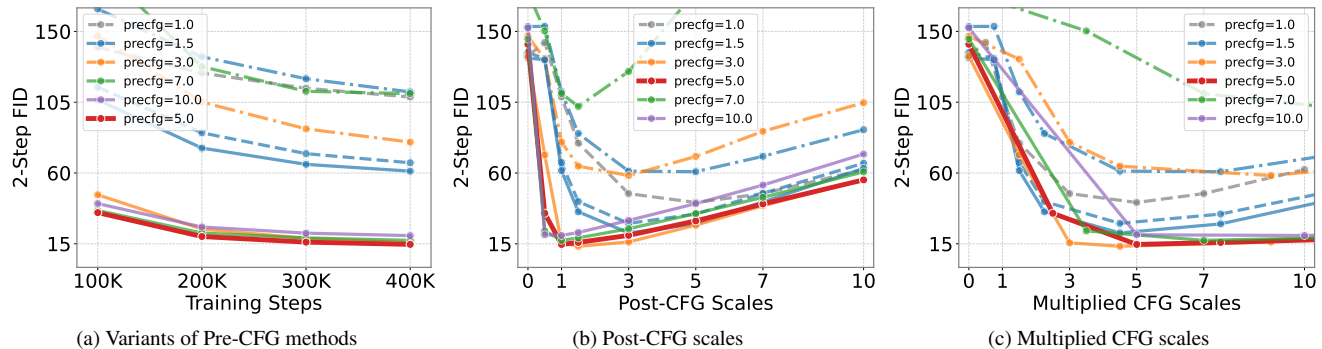


Figure 5. **Variants of Pre-CFG.** Dash, dash-dotted, and solid lines indicate iSD-U, iSD-C, and iSD-T, respectively. Each color represents a different Pre-CFG scale. Trigonometric interpolation and the JVP approximation are used by default on DiT-B/4. (on DiT-B/4a) FIDs of Pre-CFG variants over training steps. (b) FIDs of Post-CFG over guidance scales. (c) FIDs over multiplied guidance scales in Pre-CFG and Post-CFG.

particularly at $\omega = 3.0$. As we discussed earlier, applying Post-CFG on top of Pre-CFG can be viewed as applying CFG twice, resulting in a multiplied guidance scale. In this case, since iSD-U is trained at $\omega = 1.5$, the total scale becomes $4.5 = 1.5 \times 3.0$.

Similarly, iSD-T achieves its lowest FIDs around a Pre-CFG scale of $\omega = 5.0$, while Post-CFG on it underperforms even when its guidance scale is close to one. When a Pre-CFG scale of $\omega = 3.0$ is applied to iSD-T, Post-CFG outperforms at $\omega = 1.5$, corresponding to a total scale of 4.5. *These consistent observations suggest that the effective CFG scale lies near 5.0.* Based on these findings, we propose the following heuristic: (i) train vanilla iSD, (ii) measure FIDs with Post-CFG, and (iii) train iSD-T using that optimal Post-CFG scale as the Pre-CFG guidance scale.

iSD-C demonstrates optimal effective CFG scales between 7.5-10.5, distinct from the ω -unconditional settings (iSD-U and iSD-T), which lie around 4.5-5.0. However, even within its optimal range, iSD-C underperforms compared to the unconditional variants. Since recent approaches adopt ω -conditioning in distillation, further investigation into optimal scale selection for this setting remains a promising direction for future work.

Pre-CFG Scale for larger networks. For iSD-U and iSD-T, we minimally investigate the effective guidance scale for the larger network, DiT-XL/2, as shown in Tab. 9. We train DiT-XL/2 with $\omega = 5.0$ and measure quantitative metrics across Post-CFG scales (Tab. 7).

Table 7. FIDs and ISs across Post-CFG scales for DiT-XL/2 with iSD-T ($\omega = 5.0$, 600K steps).

Post-CFG ω	0.0	0.5	1.0	1.5	3.0	5.0	7.0	10.0
FID	100.89	11.19	16.96	18.26	19.33	25.83	35.59	48.50
IS	13.53	164.48	234.85	229.35	181.94	116.25	85.75	65.36

DiT-XL/2 shows better results at the Post-CFG scale of $\omega = 0.5$, suggesting an effective Pre-CFG scale near 2.5. As reported in Tab. 9, the Pre-CFG scale of $\omega = 3.0$ outperforms $\omega = 5.0$, supporting the previously introduced heuristics. This indicates that *the effective Pre-CFG scale may not generalize to larger networks*, and we report the quantitative results for the Pre-CFG scale $\omega = 3.0$ in Tab. 5. A similar trend has also been observed in prior work, where different CFG scales were used across architectures. Generalizing hyperparameters for larger networks using Muon-like optimizers [20] is left for future work.

Network-free Pre-CFG target. Since iSD-T requires $F_{t,t}^\theta(x_t; \emptyset)$ for the modified flow matching target, it incurs additional network passes at every training step. To mitigate this cost, we explored replacing it with an analytic form of the marginal velocity:

$$v_t^*(x_t) = \sum_{x_i \in X} \left[\gamma_{t,i}(x_t) \cdot \frac{x_t - x_i}{t} \right], \quad \gamma_{t,i}(x_t) = \text{softmax}_i \left(\left\{ -\frac{x_t - (1-t)x_j}{2t} \right\}_{x_j \in X} \right)$$

where $\gamma_{t,i}$ denotes the posterior responsibility, and softmax_i represents the i -th output element of the softmax function over data points. However, this formulation requires a full iteration over the dataset at every step, which becomes intractable for large datasets. To make training feasible, we approximate the marginal velocity by restricting the computation to a mini-batch $X_B \subset X$

However, under the consistency training setting, it plateaued at an FID of 343.7 and an IS of 1.0 even after 120K training steps, and we conclude that it failed to train. Under iSD-T, we observed a similar trend, with an FID of 261.1 and an IS of 1.927 at 120K steps. We hypothesize that approximating the marginal velocity at the batch level introduces larger errors than the network approximation $F_{t,t}^\theta(x_t; \emptyset)$, thereby destabilizing the training process. Extending the training-time CFG to the 1-NFE setting remains future work.

Proposal Distribution. Following prior work, we compare proposal distributions between the Log-normal [9, 23] and Beta distributions [38]. For the Log-normal distribution, we adopt the setting of MeanFlow, $\mu = -0.4$ and $\sigma = 1.0$. Under this configuration, iSD-T with $\omega = 5.0$ achieves an FID of 15.11 and an IS of 180.24, slightly underperforming Beta(0.8, 1.0).

Table 8. **Quantitative results across design choices.** The numeric entries in the header denote Post-CFG scales. Default settings follow the ablation setup (DiT-B/4, 400K training steps).

(a) Frechet Inception Distance											
Loss	Interp.	JVP	Pre-CFG ω	0.0	0.5	1.0	1.5	3.0	5.0	7.0	10.0
$\mathcal{L}_{CT}$	Linear	Exact	-	103.87	113.00	73.41	50.46	39.33	53.38	68.82	85.87
$\mathcal{L}_{CT}$	Trig	Exact	-	128.97	134.98	95.58	64.51	33.77	30.78	38.91	59.11
$\mathcal{L}_{iSD}$	Linear	Exact	-	132.36	137.38	115.60	96.30	76.26	79.86	89.88	102.06
$\mathcal{L}_{iSD}$	Trig	Exact	-	134.72	142.76	107.32	78.11	46.48	41.92	48.24	64.29
$\mathcal{L}_{CT}$	Linear	Approx	-	96.09	107.01	63.42	41.33	33.53	50.17	66.68	84.49
$\mathcal{L}_{iSD}$	Linear	Approx	-	127.07	132.37	108.57	89.15	70.73	76.06	87.46	100.01
$\mathcal{L}_{iSD}$	Trig	Approx	-	134.50	142.96	108.33	79.04	46.91	41.26	46.92	62.20
$\mathcal{L}_{iSD-U}$	Linear	Exact	1.5	122.32	123.57	73.40	55.59	53.89	72.76	86.95	99.97
$\mathcal{L}_{iSD-U}$	Trig	Exact	1.5	136.48	132.76	66.63	41.84	27.99	33.92	47.05	66.26
$\mathcal{L}_{iSD-C}$	Linear	Exact	1.5	124.69	125.28	100.47	85.78	75.32	84.40	95.58	106.94
$\mathcal{L}_{iSD-C}$	Linear	Exact	3.0	131.82	126.66	88.93	76.76	80.34	100.27	117.79	136.33
$\mathcal{L}_{iSD-C}$	Trig	Exact	1.5	153.12	153.23	111.56	85.14	61.20	60.85	70.61	87.50
$\mathcal{L}_{iSD-C}$	Trig	Exact	3.0	146.95	132.38	79.64	64.36	58.39	70.51	86.49	104.69
$\mathcal{L}_{iSD-C}$	Trig	Exact	7.0	174.91	150.28	110.49	102.30	124.54	175.54	221.82	247.85
$\mathcal{L}_{iSD-T}$	Linear	Approx	1.5	112.08	115.91	63.81	46.58	46.70	65.50	79.95	92.81
$\mathcal{L}_{iSD-T}$	Linear	Approx	3.0	108.40	81.81	27.70	26.44	39.45	56.93	68.19	79.54
$\mathcal{L}_{iSD-T}$	Linear	Approx	5.0	112.18	56.60	27.32	31.91	45.80	58.43	66.54	77.75
$\mathcal{L}_{iSD-T}$	Linear	Approx	7.0	114.42	45.37	28.61	34.30	46.54	56.35	64.28	76.59
$\mathcal{L}_{iSD-T}$	Linear	Approx	10.0	120.52	41.34	33.86	41.37	54.21	63.74	72.58	84.48
$\mathcal{L}_{iSD-T}$	Trig	Approx	1.5	133.18	131.91	61.52	35.35	21.92	27.66	41.65	62.91
$\mathcal{L}_{iSD-T}$	Trig	Approx	3.0	134.28	71.53	15.63	13.44	16.15	26.86	39.18	55.80
$\mathcal{L}_{iSD-T}$	Trig(Ours)	Approx	5.0	141.82	34.42	14.61	15.87	20.37	29.56	40.32	55.65
$\mathcal{L}_{iSD-T}$	Trig	Approx	7.0	145.19	23.22	17.21	18.78	24.56	34.25	44.52	60.72
$\mathcal{L}_{iSD-T}$	Trig	Approx	10.0	152.19	20.75	20.29	22.19	29.87	40.54	52.33	72.04
(b) Inception Score											
Loss	Interp.	JVP	Pre-CFG ω	0.0	0.5	1.0	1.5	3.0	5.0	7.0	10.0
$\mathcal{L}_{CT}$	Linear	Exact	-	11.61	11.11	19.94	30.68	39.68	29.37	22.23	16.99
$\mathcal{L}_{CT}$	Trig	Exact	-	11.10	10.83	18.75	31.94	60.81	62.27	49.07	31.40
$\mathcal{L}_{iSD}$	Linear	Exact	-	8.90	8.63	10.80	13.42	16.37	15.32	13.56	11.65
$\mathcal{L}_{iSD}$	Trig	Exact	-	10.50	9.92	15.25	22.80	37.42	38.31	32.66	23.84
$\mathcal{L}_{CT}$	Linear	Approx	-	12.50	11.78	23.92	38.45	48.40	33.31	24.36	18.16
$\mathcal{L}_{iSD}$	Linear	Approx	-	9.24	8.85	11.60	14.49	17.88	16.40	14.16	12.18
$\mathcal{L}_{iSD}$	Trig	Approx	-	10.56	9.96	14.93	22.32	37.20	39.60	33.84	25.11
$\mathcal{L}_{iSD-U}$	Linear	Exact	1.5	9.90	10.36	20.12	26.57	26.18	18.82	15.16	12.60
$\mathcal{L}_{iSD-U}$	Trig	Exact	1.5	10.33	11.43	29.84	47.81	64.99	52.43	38.11	25.94
$\mathcal{L}_{iSD-C}$	Linear	Exact	1.5	9.64	9.80	12.54	14.06	14.62	13.17	11.83	10.44
$\mathcal{L}_{iSD-C}$	Linear	Exact	3.0	8.78	9.72	13.98	15.11	13.65	10.95	9.35	8.06
$\mathcal{L}_{iSD-C}$	Trig	Exact	1.5	8.64	8.88	13.88	18.32	22.91	20.85	17.64	14.34
$\mathcal{L}_{iSD-C}$	Trig	Exact	3.0	9.03	11.19	18.84	21.40	20.53	16.87	14.24	12.03
$\mathcal{L}_{iSD-C}$	Trig	Exact	7.0	6.78	9.47	13.12	12.73	9.23	6.74	4.82	3.57
$\mathcal{L}_{iSD-T}$	Linear	Approx	1.5	10.80	11.05	23.85	32.44	31.49	21.87	17.31	14.47
$\mathcal{L}_{iSD-T}$	Linear	Approx	3.0	11.23	18.61	63.65	70.84	46.35	30.70	24.65	20.12
$\mathcal{L}_{iSD-T}$	Linear	Approx	5.0	10.79	30.42	73.67	65.97	42.05	31.02	26.35	21.80
$\mathcal{L}_{iSD-T}$	Linear	Approx	7.0	10.57	39.72	75.09	64.07	42.44	33.08	28.50	23.04
$\mathcal{L}_{iSD-T}$	Linear	Approx	10.0	10.18	43.47	63.18	51.34	34.35	27.92	23.91	19.86
$\mathcal{L}_{iSD-T}$	Trig	Approx	1.5	10.70	11.54	35.18	60.63	85.63	68.93	47.11	30.07
$\mathcal{L}_{iSD-T}$	Trig	Approx	3.0	10.65	32.80	135.51	161.59	146.40	92.51	64.00	43.48
$\mathcal{L}_{iSD-T}$	Trig(Ours)	Approx	5.0	9.86	79.12	184.11	188.96	147.41	95.08	69.11	49.04
$\mathcal{L}_{iSD-T}$	Trig	Approx	7.0	9.63	112.75	184.36	178.49	126.55	84.16	64.48	45.55
$\mathcal{L}_{iSD-T}$	Trig	Approx	10.0	9.05	127.12	163.11	151.33	101.92	70.49	53.11	36.63

Table 9. **Quantitative results over training steps** (2-NFE). The numeric entries in the header denote training steps. Reported results are medians over three random initializations, and *Std.* denotes the standard deviation at 400K steps. Hyphens in *Std.* indicate cases not measured.

(a) Frechet Inception Distance									
Loss	Interp.	JVP	Arch.	Pre-CFG ω	100K	200K	300K	400K	Std.
Baseline	Linear	Exact	DiT-B/4	-	143.82	132.03	127.37	124.75	2.32
Joint-training	Linear	Exact	DiT-B/4	-	144.34	115.57	106.95	112.13	10.08
$\mathcal{L}_{CT}$	Linear	Exact	DiT-B/4	-	100.81	85.17	77.57	73.37	1.30
$\mathcal{L}_{CT}$	Trig	Exact	DiT-B/4	-	131.41	108.29	100.52	95.58	3.40
$\mathcal{L}_{iSD}$	Linear	Exact	DiT-B/4	-	156.04	127.55	120.00	115.60	1.10
$\mathcal{L}_{iSD}$	Trig	Exact	DiT-B/4	-	146.20	123.80	115.04	107.32	4.04
$\mathcal{L}_{CT}$	Linear	Approx	DiT-B/4	-	94.55	75.21	68.10	63.42	0.17
$\mathcal{L}_{CT}$	Trig	Approx	DiT-B/4	-	223.10	227.90	228.70	Halted	-
$\mathcal{L}_{iSD}$	Linear	Approx	DiT-B/4	-	142.59	119.00	110.93	108.57	3.51
$\mathcal{L}_{iSD}$	Trig	Approx	DiT-B/4	-	139.95	123.56	113.88	108.33	-
$\mathcal{L}_{iSD-U}$	Linear	Approx	DiT-B/4	1.5	128.68	95.21	83.36	76.29	1.48
$\mathcal{L}_{iSD-U}$	Trig	Approx	DiT-B/4	1.5	114.59	85.57	72.31	66.51	0.83
$\mathcal{L}_{iSD-C}$	Linear	Approx	DiT-B/4	1.5	166.93	128.24	110.30	100.47	-
$\mathcal{L}_{iSD-C}$	Linear	Approx	DiT-B/4	3.0	200.17	120.56	98.73	88.93	-
$\mathcal{L}_{iSD-C}$	Linear	Approx	DiT-B/4	7.0	270.50	Halted	-	-	-
$\mathcal{L}_{iSD-C}$	Trig	Approx	DiT-B/4	1.5	164.34	133.90	119.94	111.56	-
$\mathcal{L}_{iSD-C}$	Trig	Approx	DiT-B/4	3.0	147.15	105.09	88.08	79.64	-
$\mathcal{L}_{iSD-C}$	Trig	Approx	DiT-B/4	7.0	192.14	127.65	111.93	110.49	-
$\mathcal{L}_{iSD-T}$	Linear	Approx	DiT-B/4	1.5	115.18	83.02	70.88	63.81	-
$\mathcal{L}_{iSD-T}$	Linear	Approx	DiT-B/4	3.0	61.37	39.82	31.90	27.70	0.40
$\mathcal{L}_{iSD-T}$	Linear	Approx	DiT-B/4	5.0	51.65	35.08	29.98	27.32	0.35
$\mathcal{L}_{iSD-T}$	Linear	Approx	DiT-B/4	7.0	51.88	36.85	31.46	28.61	-
$\mathcal{L}_{iSD-T}$	Linear	Approx	DiT-B/4	10.0	54.90	41.77	37.07	33.86	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/4	1.5	106.54	75.85	65.50	61.19	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/4	3.0	46.15	24.37	18.43	15.63	0.12
$\mathcal{L}_{iSD-T}$	Trig(Ours)	Approx	DiT-B/4	5.0	34.81	19.60	16.05	14.61	0.05
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/4	7.0	36.17	21.83	18.72	17.17	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/4	10.0	40.54	25.69	21.75	20.18	-
MeanFlow	Linear	Exact	DiT-B/4	3.0	41.29	24.37	18.20	14.83	2.00
MeanFlow	Trig	Exact	DiT-B/4	3.0	46.48	21.61	15.59	12.88	3.11
$\mathcal{L}_{iSD-U}$	Trig	Approx	DiT-B/2	1.5	87.47	55.21	42.48	36.05	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/2	3.0	36.40	19.20	13.83	11.92	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/2	5.0	31.00	18.96	16.75	15.93	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-L/2	3.0	26.02	14.86	12.11	11.17	-
$\mathcal{L}_{iSD-U}$	Trig	Approx	DiT-XL/2	1.5	87.47	55.21	42.48	36.05	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2	3.0	24.69	14.33	11.82	11.08	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2	5.0	28.30	18.89	17.33	17.03	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2 [†]	3.0	31.35	16.10	12.21	10.80	-

Loss	Interp.	JVP	Arch.	Pre-CFG ω	500K	600K	700K	800K
$\mathcal{L}_{iSD-U}$	Trig	Approx	DiT-XL/2	1.5	30.87	27.56	24.80	23.40
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2	3.0	10.66	10.48	10.29	10.30
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2	5.0	16.87	16.85	16.96	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2 [†]	3.0	10.22	9.78	9.81	9.70

(b) Inception Score

Loss	Interp.	JVP	Arch.	Pre-CFG ω	100K	200K	300K	400K	Std.
Baseline	Linear	Exact	DiT-B/4	-	7.51	8.37	8.86	9.12	0.37
Joint-training	Linear	Exact	DiT-B/4	-	9.65	10.65	10.79	9.96	2.90
$\mathcal{L}_{CT}$	Linear	Exact	DiT-B/4	-	13.34	16.37	18.40	19.72	0.50
$\mathcal{L}_{CT}$	Trig	Exact	DiT-B/4	-	11.78	15.42	17.35	18.75	0.98
$\mathcal{L}_{iSD}$	Linear	Exact	DiT-B/4	-	7.26	9.55	10.38	10.80	0.08
$\mathcal{L}_{iSD}$	Trig	Exact	DiT-B/4	-	9.37	12.31	13.93	15.25	0.70
$\mathcal{L}_{CT}$	Linear	Approx	DiT-B/4	-	14.91	19.29	21.79	23.92	0.11
$\mathcal{L}_{CT}$	Trig	Approx	DiT-B/4	-	3.76	3.85	3.89	Halted	-
$\mathcal{L}_{iSD}$	Linear	Approx	DiT-B/4	-	8.74	10.80	11.61	11.60	0.39
$\mathcal{L}_{iSD}$	Trig	Approx	DiT-B/4	-	10.02	11.80	13.68	14.93	-
$\mathcal{L}_{iSD-U}$	Linear	Approx	DiT-B/4	1.5	10.00	15.01	17.21	19.02	0.45
$\mathcal{L}_{iSD-U}$	Trig	Approx	DiT-B/4	1.5	13.38	20.84	26.33	29.62	0.52
$\mathcal{L}_{iSD-C}$	Linear	Approx	DiT-B/4	1.5	6.45	9.33	11.18	12.54	-
$\mathcal{L}_{iSD-C}$	Linear	Approx	DiT-B/4	3.0	4.88	10.16	12.55	13.98	-
$\mathcal{L}_{iSD-C}$	Linear	Approx	DiT-B/4	7.0	1.83	Halted	-	-	-
$\mathcal{L}_{iSD-C}$	Trig	Approx	DiT-B/4	1.5	8.04	10.95	12.58	13.88	-
$\mathcal{L}_{iSD-C}$	Trig	Approx	DiT-B/4	3.0	9.14	14.30	17.17	18.84	-
$\mathcal{L}_{iSD-C}$	Trig	Approx	DiT-B/4	7.0	4.69	10.79	12.42	13.12	-
$\mathcal{L}_{iSD-T}$	Linear	Approx	DiT-B/4	1.5	11.87	17.14	20.84	23.85	-
$\mathcal{L}_{iSD-T}$	Linear	Approx	DiT-B/4	3.0	25.05	41.33	53.72	63.65	0.99
$\mathcal{L}_{iSD-T}$	Linear	Approx	DiT-B/4	5.0	30.66	50.89	63.39	73.67	1.88
$\mathcal{L}_{iSD-T}$	Linear	Approx	DiT-B/4	7.0	31.42	50.69	63.81	75.09	-
$\mathcal{L}_{iSD-T}$	Linear	Approx	DiT-B/4	10.0	28.87	42.85	53.79	63.18	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/4	1.5	15.68	25.63	31.76	35.41	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/4	3.0	39.41	84.84	114.06	135.51	0.19
$\mathcal{L}_{iSD-T}$	Trig(Ours)	Approx	DiT-B/4	5.0	53.61	118.14	158.99	184.11	0.50
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/4	7.0	52.79	117.45	157.02	186.79	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/4	10.0	46.28	100.12	137.87	164.59	-
MeanFlow	Linear	Exact	DiT-B/4	3.0	40.01	75.01	104.54	126.99	24.39
MeanFlow	Trig	Exact	DiT-B/4	3.0	40.18	98.86	137.02	162.85	22.16
$\mathcal{L}_{iSD-U}$	Trig	Approx	DiT-B/2	1.5	16.28	25.70	34.44	42.48	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/2	3.0	44.96	93.25	132.21	156.90	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-B/2	5.0	56.63	122.93	153.30	170.99	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-L/2	3.0	61.29	122.22	159.68	178.42	-
$\mathcal{L}_{iSD-U}$	Trig	Approx	DiT-XL/2	1.5	16.28	25.70	34.44	42.48	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2	3.0	64.95	127.97	165.02	185.29	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2	5.0	66.44	140.28	179.87	201.29	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2 [†]	3.0	48.17	108.24	151.14	181.05	-

Loss	Interp.	JVP	Arch.	Pre-CFG ω	500K	600K	700K	800K
$\mathcal{L}_{iSD-U}$	Trig	Approx	DiT-XL/2	1.5	51.00	57.25	64.07	68.70
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2	3.0	199.70	212.66	225.43	233.06
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2	5.0	220.07	233.32	234.85	-
$\mathcal{L}_{iSD-T}$	Trig	Approx	DiT-XL/2 [†]	3.0	202.50	220.72	234.24	246.85

D. Qualitative Results

1300



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Figure 6. One-step samples from the vanilla iSD on CIFAR-10 (FID 3.64)

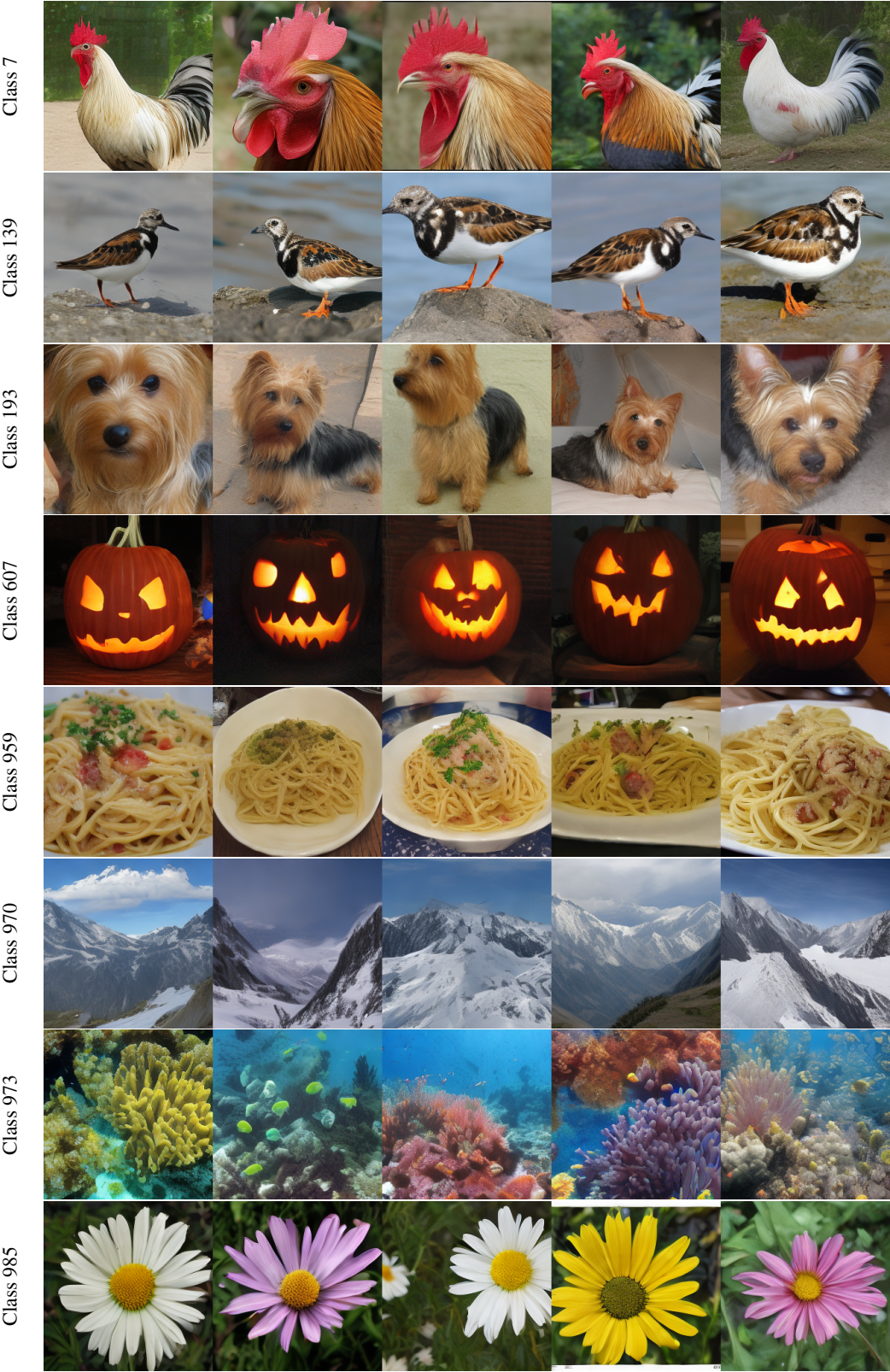


Figure 7. Class-level samples generated by iSD-T with two-step sampling on ImageNet 256 × 256

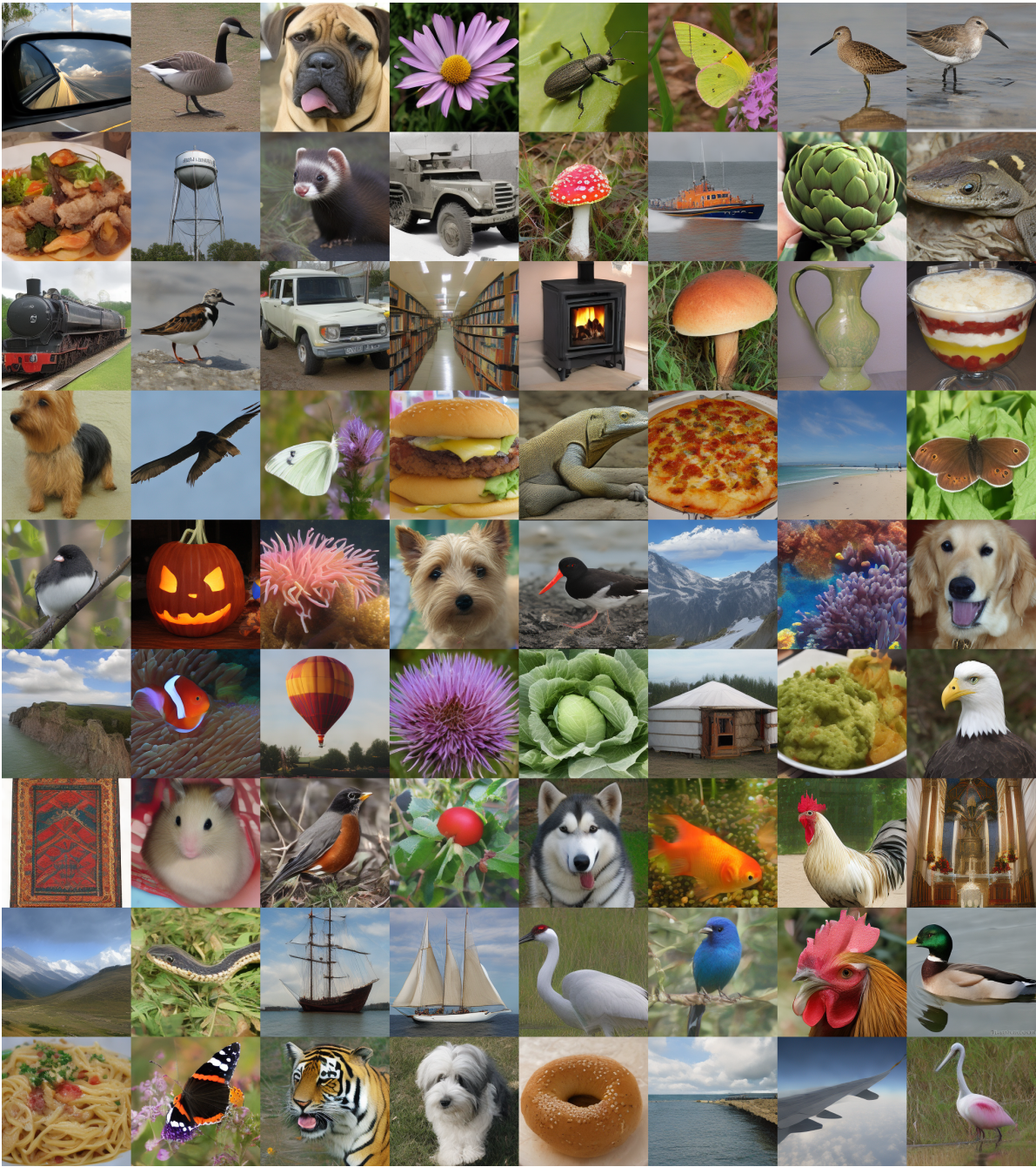


Figure 8. Two-step samples from the iSD-T on ImageNet 256 × 256